

Invariant-projection counterexamples for every irrational rotation

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Abstract

For every irrational angle, we construct a continuous nonnegative circle weight with exactly one zero and logarithmic integral $-\infty$ whose weighted rotation has no nontrivial invariant projection in the associated hyperfinite type II_1 factor. This answers negatively the question of Zhu, Fang, and Shi, with the angle prescribed in advance. The resulting operator is nonzero and norm-quasinilpotent.

1 Introduction

Let $\theta \in (0, 1)$ be irrational. In the tracial representation let R_θ be the von Neumann algebra generated by unitaries U, V with

$$VU = e^{2\pi i\theta}UV, \quad \tau(U^m V^n) = 0 \quad ((m, n) \in \mathbb{Z}^2 \setminus \{(0, 0)\}), \quad \tau(1) = 1.$$

This is the hyperfinite factor of type II_1 ; the concrete representation used below is the group-measure-space representation of irrational rotation. For a continuous function f on the unit circle $\mathbb{T}$, write $T_f = Uf(V)$. A closed invariant subspace is *affiliated with* R_θ when its orthogonal projection belongs to R_θ . A nontrivial such subspace is therefore equivalent to a projection $p \in R_\theta$ satisfying

$$0 < \tau(p) < 1, \quad (1 - p)T_f p = 0.$$

There is no requirement that p commute with T_f .

Let m denote normalized Haar measure on $\mathbb{T}$. The Fuglede–Kadison determinant of the scalar weight is

$$\Delta(f(V)) = \exp \left(\int_{\mathbb{T}} \log |f(z)| \, dm(z) \right),$$

with $\log 0 = -\infty$ and $\exp(-\infty) = 0$. Zhu, Fang, and Shi asked whether T_f must have a nontrivial affiliated invariant subspace if f is nonzero almost everywhere, $|f|$ is nonconstant, and this determinant is zero [14, p. 264, Question after Proposition 5.7]. We answer their question negatively for every prescribed irrational angle.

Theorem 1.1. *For every irrational $\theta \in (0, 1)$ there exists a continuous function $f : \mathbb{T} \rightarrow [0, 1]$ such that*

$$f^{-1}(\{0\}) = \{1\}, \quad \int_{\mathbb{T}} \log f \, dm = -\infty,$$

and every projection $p \in R_\theta$ satisfying $(1 - p)Uf(V)p = 0$ is either 0 or 1.

The angle is fixed before the weight is constructed; no Diophantine condition is imposed. The conclusion concerns projections in the specified containing factor. Arbitrary invariant subspaces of its Hilbert-space representation need not have their projections in that factor. Corollary 6.2 also shows directly that the constructed operator is nonzero and norm-quasinilpotent.

1.1 Hyperinvariant subspaces and commutants

For a bounded operator T on a complex Hilbert space $\mathcal{H}$, write $\{T\}' = \{A \in B(\mathcal{H}) : AT = TA\}$. A closed subspace is *hyperinvariant* for T if every member of $\{T\}'$ leaves it invariant. An operator algebra is *transitive* if its only common closed invariant subspaces are 0 and $\mathcal{H}$.

Lemma 1.2 (Commutant algebra). *If $T \in B(\mathcal{H})$ is nonscalar, then $\{T\}'$ is a proper, strongly operator closed, unital complex algebra. It is transitive exactly when T has no nontrivial hyperinvariant subspace.*

Proof. The commutation equation is preserved by addition, multiplication and scalar multiplication, and the identity commutes with T . If $A_\lambda \rightarrow A$ strongly and $A_\lambda T = TA_\lambda$, then for every $\zeta \in \mathcal{H}$ both sides converge, giving $AT\zeta = TA\zeta$. Thus the algebra is strongly closed. If every bounded operator commuted with T , then every rank-one orthogonal projection would commute with T , so every nonzero vector would be an eigenvector. Applying this to $x, y, x + y$ for linearly independent vectors shows that the eigenvalues are equal; hence T would be scalar. Properness follows. The last assertion is precisely the two definitions. $\square$

The bridge to the containing von Neumann algebra is the elementary observation recorded in Dykema [4, Proposition 2.1].

Lemma 1.3. *Let $\mathcal{M} \subset B(\mathcal{H})$ be a von Neumann algebra and $T \in \mathcal{M}$. The orthogonal projection onto any hyperinvariant subspace for T belongs to $\mathcal{M}$ and is invariant for T .*

Proof. Every unitary $W \in \mathcal{M}'$ and its inverse commute with T . Hyperinvariance therefore makes the subspace reducing for W . Its orthogonal projection commutes with all such unitaries and hence belongs to $\mathcal{M}'' = \mathcal{M}$. Since $T \in \{T\}'$, its range is invariant under T . $\square$

Corollary 1.4. *Every separable infinite-dimensional complex Hilbert space carries a nonzero norm-quasinilpotent operator with no nontrivial hyperinvariant subspace. Its commutant is a proper strongly closed transitive unital algebra.*

Proof. The tracial representation of R_θ is separable and infinite-dimensional. Apply Theorem 1.1, Corollary 6.2, and Lemma 1.3 there. A nonzero quasinilpotent operator is nonscalar, so Lemma 1.2 applies. Unitary equivalence transfers these properties to any separable infinite-dimensional complex Hilbert space. $\square$

These conclusions do not exclude ordinary invariant subspaces in the representation. A separate companion gives a direct 2-adic weighted shift with a transitive commutant, including an explicit norm estimate, backward intertwiners and a description of its polar conjugates [11]. It uses only Lemmas 1.2 and 3.1 from this paper. The present proof uses no result from that companion.

Remark 1.5. Corollary 1.4 contradicts the hyperinvariant-subspace assertion of Chō et al. [3, Corollary 2.8, p. 801] at $\mu = 1$: for the nonzero quasinilpotent T just constructed and any conjugation C , their stated hypotheses apply to $I + T$, but $\{I + T\}' = \{T\}'$, so $I + T$ has no nontrivial hyperinvariant subspace. Only that hyperinvariant-subspace assertion is at issue here.

1.2 Prior work

The determinant of Fuglede and Kadison [8] and Brown's spectral distribution [2] form a major route to invariant subspaces in finite factors. Haagerup and Schultz associate an affiliated hyperinvariant projection to each Borel subset of the plane, with trace equal to its Brown mass [9, Theorem 1.1]. Thus a Brown measure that is not a point mass gives a

nontrivial projection. For a point-mass Brown measure, the trace formula gives only trivial projections.

Point-mass Brown measure can nevertheless coexist with many invariant subspaces. Dykema and Haagerup construct a one-parameter family of nontrivial affiliated hyperinvariant subspaces for the quasinilpotent DT-operator [5, Theorems 2.2 and 2.9]. Thus degeneration of Brown measure to a point mass does not itself exclude invariant projections. Tucci also constructed quasinilpotent generators of the hyperfinite factor with nontrivial affiliated invariant projections [13, Theorems 3.1 and 3.3 and the discussion following Theorem 3.3]. These examples further distinguish quasinilpotence from the absence of factor-invariant projections.

For weighted irrational rotations, the measurable-action theorem of Dykema and Schultz identifies the Brown measure as uniform measure on the circle whose radius is the determinant [6, Theorem 5.4]. Zhu, Fang, and Shi give a direct proof, explicitly crediting that earlier result [14, Theorem 5.3]. At determinant zero this measure is the point mass at zero, so the Brown-measure projections are trivial.

Continuous weights with one zero and determinant zero were already exhibited by Zhu, Fang, and Shi [14, pp. 262–263]. The additional conclusion of Theorem 1.1 is the exclusion of every nontrivial invariant projection for a suitable weight at each prescribed angle. Zhu, Fang, and Shi’s spectral-radius formula [14, Theorem 2.3] implies quasinilpotence in this case. We give an elementary proof of this special case, using only uniform averaging of continuous functions under irrational rotation. Their positive-determinant invariant-subspace result [14, Corollary 5.5] has a hypothesis complementary to the one studied here.

1.3 Proof strategy

The obstruction begins with a measurable weight. Suppose its weighted rotation had a nontrivial invariant projection. In the shift representation, this projection becomes a field of projections. The finite trace and ergodicity force its central diagonal entry to lie strictly between zero and one almost everywhere. Thus both the projection and its complement have vectors with a nonzero central coordinate.

Invariance imposes weighted orthogonality identities on these vectors. We first obtain the identities along rotation orbits and then pass them to suitably coupled pairs of points. The weight has drops that are rare enough to leave a chosen finite window under control, but strong enough to suppress contributions outside that window. The central coordinates force at least one finite convolution coefficient to stay away from zero. The suppressed tail cannot cancel it, contradicting the exact orthogonality identity. Conditioning separately on each endpoint controls that tail even though it has infinitely many terms.

The measurable weight is then replaced by a continuous weight with one zero. Smoothing and moving the logarithmic drops changes their sum by a constant plus a measurable coboundary. This relation supplies an intertwining multiplier, possibly unbounded. A domain and trace argument shows that an invariant projection for the continuous operator would produce one for the measurable operator with exactly the same trace. This transfers the obstruction to the desired continuous weight.

The construction realizing this scheme is as follows. Use the additive circle $X = \mathbb{R}/\mathbb{Z}$, with Haar probability μ , and write $\sigma x = x + \theta$. Write $\{t\} \in [0, 1)$ for fractional part. We construct divisible integers $1 = q_0 \mid q_1 \mid q_2 \mid \cdots$ and positive costs c_i . The measurable model is the weight

$$g_*(x) = \sum_{i \geq 1} c_i [\{q_i x\} + \{q_i \theta\}], \quad f_*(x) = e^{-g_*(x)}.$$

The series is finite almost everywhere although its integral diverges. Large divisible frequencies allow its trial phases to be realized at the fixed angle θ , while finitely many sampling tests

approximate independent Haar coordinates. The circle characters themselves are dependent. A mixed-radix expansion supplies independent digits, from which we build the ordered couplings with controlled marginal densities. The prescribed-marginal viewpoint belongs to the classical coupling theory of Strassen and Edwards [7, 12]; here we also need approximation by positive orbit pairs.

Section 2 fixes the frequencies and sampling powers. Sections 3 and 4 obtain the projection identities on the ordered pairs. Section 5 proves and applies the convolution obstruction, and Section 6 constructs the continuous weight and proves the trace-preserving transfer.

Appendix A gives a complete alternative realization on an infinite product of circles. Independent coordinates yield a finite-coordinate orbit-sampling theorem and a second application of the same convolution obstruction. That model also separates the atomic algebras generated by finitely many polar conjugates from the diffuse algebra generated by the full orbit. The prescribed-angle circle proof is complete before the appendix.

All identities involving measurable fields hold on appropriate conull sets. When integer iterates are used, we intersect the countably many translates of these sets. The constants implicit in $O(\cdot)$ below are absolute unless a fixed construction stage is specified. Every new ingredient needed for the proof is established in the paper.

2 Sampling at a prescribed irrational angle

Fix $X = \mathbb{R}/\mathbb{Z}$ with Haar probability μ , and put $\sigma x = x + \theta$, where $\theta \in (0, 1)$ is a fixed irrational number. We write $\{t\} \in [0, 1)$ for fractional part. Our immediate aim is to construct rare logarithmic drops whose characters behave almost like independent coordinates in finitely many sampling tests. The frequencies will be divisible integers; the angle θ will never be changed.

We use the following elementary facts about irrational rotations. For each nonzero integer a , the average of $e^{2\pi i a n \theta}$ over $1 \leq n \leq N$ tends to zero, by the formula for a geometric sum. Trigonometric approximation gives equidistribution against continuous functions, and continuous upper and lower bounds give it for intervals. The same argument gives convergence uniform in the starting point for continuous functions. In particular, every positive orbit is dense. These facts also hold with θ replaced by any positive integer multiple of θ .

For an integer $m \geq 200$, put $h = m^{-3/5}$ and $\Delta(z) = \min(\{z\}, 1 - \{z\})$. Define on two circles

$$w_m(z, v) = \mathbf{1}_{\{\Delta(z) \geq m^{-2}\}} \frac{\mathbf{1}_{\{0 < v < h\Delta(z)\}}}{h\Delta(z)}, \quad w_m^{(1)}(z, v) = w_m(z, v), \quad w_m^{(2)}(z, v) = w_m(z - v, v), \quad (2.1)$$

where v is represented in $[0, 1)$ and the expression is zero when the cutoff fails. The first kernel forces a small positive move with no wrap around zero. The second is its density viewed from the endpoint.

The kernels have almost unit mass at either endpoint. We next control their second moments and the correlations produced by two separated integer powers. These estimates will let us average products over a whole block of coordinates. The scalar lemmas also serve the independent-coordinate construction in Appendix A.

Lemma 2.1 (Scalar sampling estimates). *Let $m \geq 200$ be an integer, put $h = m^{-3/5}$, and let $u = w_m^{(r)}$ for either $r = 1$ or $r = 2$. Write $\bar{u}(s) = \int_0^1 u(s, v) dv$. Uniformly in s , the supremum*

plus total variation of $v \mapsto u(s, v)$ is $O(m^{13/5})$, hence $O(m^{10})$, and

$$\begin{aligned} \|\bar{u}\|_\infty &\leq 2, & \int_0^1 \bar{u} &= 1 - 2m^{-2}, \\ (1 - 2m^{-2})^2 &\leq \int_0^1 \bar{u}^2 \leq 1 + \frac{h^2}{(1-h)^2}, & \iint_{[0,1]^2} u^2 &= \frac{2}{h} \log \frac{m^2}{2}. \end{aligned} \quad (2.2)$$

For the sampling estimate let $0 < N_0 \leq N \leq m^2$, $Q = m^{(7/10)m}$ and $H = m^{300}$. Let λ be uniform on $[1/N_0, 2/N_0]$. If $n \in [Q/2, Q] \cap \mathbb{N}$, the density p_n of $\{n\lambda\}$ satisfies

$$|p_n - 1| \leq N_0/n \quad \text{almost everywhere.} \quad (2.3)$$

If two distinct such powers n, n' satisfy

$$|vn + v'n'| > Nm^{1000} \quad (0 < |v| + |v'|, |v|, |v'| \leq H, v, v' \in \mathbb{Z}),$$

then, for intervals $I, J \subset [0, 1)$,

$$|\mathbb{P}(\{n\lambda\} \in I, \{n'\lambda\} \in J) - |I||J|| = O(m^{-80}). \quad (2.4)$$

Consequently, for complex functions f, g of bounded variation on $[0, 1]$, with $B(f) = \|f\|_\infty + \text{Var}(f)$,

$$\left| \mathbb{E}[f(\{n\lambda\})g(\{n'\lambda\})] - \left(\int_0^1 f \right) \left(\int_0^1 g \right) \right| \leq Cm^{-80}B(f)B(g). \quad (2.5)$$

All implicit constants are absolute, for sufficiently large m , and uniform in N_0, N, n, n' and the intervals or functions.

Proof. For the first orientation the function of v is a constant height at most $m^{13/5}$ on an interval, so the variation bound is immediate. For the second, nonzero values require $v < s$ in representatives $s, v \in [0, 1)$: if $v > s$, then $\Delta(s - v) \leq v - s < v$, whereas $h < 1$. On $0 < v < s$, the conditions $\Delta(s - v) \geq m^{-2}$ and $v < h\Delta(s - v)$ describe an interval. The reciprocal of $\Delta(s - v)$ has at most two monotone pieces there, with the same height bound. This proves the asserted uniform bound in both orientations.

In the first orientation $\bar{u}$ is the cutoff indicator, and

$$\iint u^2 = \frac{2}{h} \int_{m^{-2}}^{1/2} \frac{ds}{s} = \frac{2}{h} \log(m^2/2).$$

The change $(s, v) \mapsto (s - v, v)$ preserves double integrals, giving the same total mass and square integral for the second orientation. To bound its averaged density, first omit the cutoff. The resulting density d is the law of $z + t\Delta(z)$ for independent uniform $z \in [0, 1)$ and $t \in (0, h)$. For fixed t this is an increasing piecewise-linear bijection of $[0, 1]$, with slopes $1 + t$ and $1 - t$. Thus $\int d = 1$ and $|d - 1| \leq h/(1 - h)$, whence

$$\int d^2 = 1 + \int (d - 1)^2 \leq 1 + \frac{h^2}{(1 - h)^2}.$$

Restoring the cutoff only decreases this nonnegative density; also $d \leq (1 - h)^{-1} \leq 2$. These observations give the upper bounds in (2.2), while Cauchy–Schwarz gives the lower bound from its exact mass. The cancellation of the linear term here is essential when multiplying estimates over m coordinates.

Periodizing a uniform interval of length n/N_0 gives (2.3); in particular $p_n \leq 2$ for large m . For the separated powers and any permitted nonzero frequency pair, direct integration on $[1/N_0, 2/N_0]$ gives

$$\left| \mathbb{E} e^{2\pi i(vn+v'n')\lambda} \right| \leq \frac{N_0}{\pi|vn+v'n'|} \leq \frac{1}{\pi} m^{-1000}.$$

To pass from characters to rectangles, convolve each interval indicator with the Fejér kernel

$$F_H(s) = \frac{1}{H} \left(\frac{\sin(\pi H s)}{\sin(\pi s)} \right)^2.$$

Expanding the squared geometric sum shows that F_H is nonnegative, has mass one, and has frequencies of magnitude below H . The convolved indicators lie in $[0, 1]$, preserve their means, and have Fourier coefficients of modulus at most one. Outside strips of width $\eta = m^{-80}$ about interval endpoints, their error is $O((H\eta)^{-1})$, by integrating the kernel tail. The marginal bound $p_n, p_{n'} \leq 2$ shows that replacing both indicators in a product costs $O(\eta + (H\eta)^{-1})$, both for the actual joint law and for Haar product measure. The remaining Fourier error is $O(H^2 m^{-1000})$. This proves (2.4); no joint density, or independence of the sampled powers, is used.

Finally, a bounded-variation function is, up to a null set, a constant plus an integral of indicators of intervals $[a, 1)$ against its complex Stieltjes measure. The constant and measure have total size at most $B(f)$; the same holds for g . Apply (2.4) to the resulting products of interval indicators, allowing the full interval. Integration against total variation gives (2.5). Changes at the countably many possible discontinuities do not matter, because the marginals have densities. $\square$

Before choosing the block scales, we record the finite-coefficient bound that will determine how much attenuation each block must supply.

Lemma 2.2 (A nonzero finite convolution). *For every integer $K \geq 0$ and $D \geq \delta > 0$ there is $\varepsilon = \varepsilon(K, D, \delta) > 0$ such that complex Laurent polynomials A, B supported in $[-K, K]$, with coefficients of modulus at most D and $|A_0|, |B_0| \geq \delta$, satisfy*

$$\max_{0 \leq n \leq 2K} |(AB)_{-n}| \geq \varepsilon.$$

Proof. The admissible polynomial pairs form a nonempty compact set. For each pair, the first nonzero exponents of A and B are at most zero, and their sum belongs to $[-2K, 0]$. The corresponding product coefficient is nonzero. The continuous displayed maximum therefore has a positive minimum on the compact set. $\square$

We now choose the parameters for the fixed circle rotation. First prepare data for the later continuity argument. Let O_j be the open arc of radius $1/(10j)$ about zero. Choose a continuous nonnegative bump supported in O_j and positive on a smaller arc. Finitely many of its translates $x \mapsto \psi(\sigma^n x)$, with $0 \leq n \leq h_j^*$, have a strictly positive sum on X , by density and compactness. Normalize by this sum to obtain continuous functions $\phi_{j,n} \geq 0$ such that

$$\sum_{n=0}^{h_j^*} \phi_{j,n} = 1, \quad \sigma^n(\text{supp } \phi_{j,n}) \subset O_j. \quad (2.6)$$

All h_j^* and $\phi_{j,n}$ are fixed before the construction below.

The parameters below are grouped into consecutive blocks of m_j coordinates. In block j , K_j is the cutoff for a finite convolution, C_j is the cost of each logarithmic drop, and L_j is the

attenuation needed to make the omitted tail smaller than a finite polynomial coefficient. The total phase increment in the block is of order $1/K_j$. The sampling horizon Q_j lies well below the next cutoff K_{j+1} , so sampling moves can be controlled in every future block. The two widths a_j, b_j will provide crossings in complementary ranges of the tail index. The proposition states these requirements together; its proof shows that they can all be chosen successively.

Proposition 2.3. *For the fixed θ there are positive integers m_j, K_j, G_j, q_i , positive real numbers L_j, C_j, ε_j , and consecutive blocks $J_j = \{d_{j-1} + 1, \dots, d_j\}$ of size m_j , with $d_0 = 0$, having the following properties. Set*

$$\begin{aligned} q_0 &= 1, & u_i(x) &= \{q_i x\}, & \alpha_i &= \{q_i \theta\}, \\ \kappa_j &= (100j)^{-1}, & N_j &= m_j K_j, \\ a_j &= G_j/m_j, & b_j &= m_j^{-19/20}, & Q_j &= m_j^{(7/10)m_j}. \end{aligned}$$

The frequencies are strictly increasing and divisible, and

$$\frac{1}{N_j} < \alpha_i < \frac{2}{N_j}, \quad q_i \mid q_{i+1}, \quad \frac{q_{i+1}}{q_i} \geq N_j \quad (i \in J_j). \quad (2.7)$$

For every integer k with $N_j/4 \leq |k| \leq m_j^{(4/5)m_j}$, at least $m_j/20$ indices in J_j satisfy

$$1 - \{k\alpha_i\} \in [b_j, 1 - a_j]. \quad (2.8)$$

If $G_j K_j \leq |k| \leq N_j/4$, every index in J_j has $1 - \{k\alpha_i\} \in [a_j, 1 - a_j]$.

The initial values are $K_1 = 100, G_1 = 1$. The numbers $L_j \geq 1$ are nondecreasing, $C_j = \max(L_j, K_j^2)$, and the sizes satisfy

$$\begin{aligned} m_j &\geq \max(K_j, 2^j, 4G_j, 200), & K_{j+1} &= \lceil m_j^{(3/4)m_j} \rceil > 2K_j, \\ G_{j+1} K_{j+1} &< m_j^{(4/5)m_j}, & \kappa_j m_j^{1/40} &\geq L_j, & \kappa_{j+1} \sqrt{G_{j+1}} &\geq L_j, \\ h_{j+1}^*/K_{j+1} &\leq 2^{-j}. \end{aligned} \quad (2.9)$$

Put $D_j = \exp((K_j + 1) \sum_{s < j} C_s m_s)$. If A, B are Laurent polynomials supported in $[-K_j, K_j]$, every coefficient has modulus at most D_j , and $|A_0|, |B_0| \geq 1/j$, then

$$\max_{0 \leq n \leq 2K_j} |(AB)_{-n}| \geq \varepsilon_j, \quad (2K_j + 1)3e^{-L_j/10000} \leq \varepsilon_j/j. \quad (2.10)$$

Finally put $p_* = d_{j-1}$ and $\ell_j = ((j+1)^2(1+p_*)q_{p_*})^{-1}$. For every $t = r/q_{p_*}$ with $r \in \mathbb{Z}$, $0 \leq r < q_{p_*}$, there is a finite nonempty set $\mathcal{N}_{j,t}$ of positive integers in $[Q_j/2, Q_j]$, with $\{n\theta - t\} \in (0, \ell_j)$, such that for $r' = 1, 2$,

$$\left\| \frac{1}{|\mathcal{N}_{j,t}|} \sum_{n \in \mathcal{N}_{j,t}} \prod_{i \in J_j} w_{m_j}^{(r')}(u_i(x), \{n\alpha_i\}) - 1 \right\|_{L^2(\mu)} < \frac{1}{j}. \quad (2.11)$$

The proof separates two choices. We first choose independent trial phases to obtain the finite probabilistic tests. Only afterwards do we realize them, to sufficient accuracy, by characters of the prescribed rotation. Large gaps between successive frequencies also make those characters converge in distribution to independent Haar variables.

Proof. Suppose K_j, G_j and all earlier blocks are fixed. Apply Lemma 2.2 with $K = K_j$, $D = D_j$ and $\delta = 1/j$ to choose ε_j . Choose $L_j \geq \max(1, L_{j-1})$ to satisfy the second inequality in (2.10) (omit L_{j-1} when $j = 1$), set $C_j = \max(L_j, K_j^2)$, and then choose an integer G_{j+1} with $\kappa_{j+1} \sqrt{G_{j+1}} \geq L_j$. All requirements of (2.9) now hold for sufficiently large $m = m_j$. In what follows earlier data remain fixed while $m \rightarrow \infty$; in particular p_*, q_{p_*}, ℓ_j and the finite list of shifts t are fixed. Write $N = mK_j$, $Q = m^{(7/10)m}$, $a = G_j/m$, and $b = m^{-19/20}$.

Separated sampling powers. Equidistribution gives at least a fixed positive multiple of $\ell_j Q$ integers $n \in [Q/2, Q]$ with $\{n\theta - t\} \in (0, \ell_j)$ for each required t , once m is sufficiently large. Set $H = m^{300}$ and $R = Nm^{1000}$. We can select at least $m^{0.69m}$ such powers, separately for every t , so that distinct selected powers obey

$$|vn + v'n'| > R \quad (|v|, |v'| \leq H, (v, v') \neq (0, 0), v, v' \in \mathbb{Z}). \quad (2.12)$$

Indeed, if $v' \neq 0$, each previously chosen n excludes at most $2R + 2$ integers n' for each frequency pair (v, v') . There are at most $(2H + 1)^2$ pairs, and $N \leq m^2$. A greedy selection thus loses only a polynomial factor in m from the available number. Since $Q = m^{0.70m}$, it eventually leaves at least $m^{0.69m}$ powers. If $v' = 0$ and $v \neq 0$, then $|vn| \geq Q/2 > R$ for large m , so this case causes no exclusion. Fix the resulting sets $\mathcal{N}_{j,t}$.

Trial phases and the spread condition. For $i \in J_j$, take independent uniform $\lambda_i \in [1/N, 2/N]$. For each integer k in the range of (2.8), the law of $\{k\lambda_i\}$ has density at most 5: it is the periodization of a uniform interval of length $|k|/N \geq 1/4$, whose density is at most $1 + N/|k|$. The probability that an index fails the strict interior version of (2.8) is therefore at most $10b$. If fewer than $m/20$ indices succeed, more than $19m/20$ fail. Independence and a union bound give failure probability at most $2^m(10b)^{(19/20)m}$. A further union over both signs of k , up to $m^{(4/5)m}$, tends to zero: the coefficient of $m \log m$ in its logarithm is $4/5 - (19/20)^2 < 0$. The lower range in the proposition requires no probabilistic choice. When $G_j K_j \leq |k| \leq N/4$, every trial phase has $a \leq |k|\lambda_i \leq 1/2$; the same observation will hold for the eventual α_i .

One-coordinate and two-power estimates. Fix an orientation, put $u = w_m^{(r')}$, and set $\bar{u}(s) = \int_0^1 u(s, v) dv$ and $A_* = \int_0^1 \bar{u}(s)^2 ds$. Lemma 2.1, with $N_0 = N$, gives $\|\bar{u}\|_\infty \leq 2$, $(1 - 2m^{-2})^2 \leq A_* \leq 1 + 4h^2$ and $\iint u^2 \leq 4h^{-1} \log m$, where $h = m^{-3/5}$. Its bounded-variation estimate also gives, uniformly in s ,

$$\mathbb{E}[u(s, \{n\lambda_i\})u(s, \{n'\lambda_i\})] = \bar{u}(s)^2 + O(m^{-50}) \quad (n \neq n'). \quad (2.13)$$

Indeed, the error from (2.5) is $O(m^{-60})$. The one-power density estimate (2.3) likewise gives $\mathbb{E}[u(s, \{n\lambda_i\})\bar{u}(s)] = \bar{u}(s)^2 + O(m^{-50})$, since $n \geq Q/2$.

The product sampling test. We now have one-coordinate and two-power estimates. Their role is to control the second moment of an average of products over a whole block; the diagonal terms will be absorbed by the large number of sampling powers, while the distinct-power terms approach the ideal product law. For independent Haar variables $z = (z_i)_{i \in J_j}$, and a fixed power set of size P , put

$$Z(z) = \frac{1}{P} \sum_{n \in \mathcal{N}_{j,t}} \prod_{i \in J_j} u(z_i, \{n\lambda_i\}), \quad Z^{\text{id}}(z) = \prod_{i \in J_j} \bar{u}(z_i).$$

By independence of the trial phases over i , the distinct-power terms of $\mathbb{E}\|Z\|_2^2$, and the terms of $\mathbb{E}\langle Z, Z^{\text{id}} \rangle$, are products of m factors $A_* + O(m^{-50})$. They equal $A_*^m + O(m^{-49})$ by telescoping, because $(1 + 4h^2 + O(m^{-50}))^m$ stays bounded. The equal-power terms are each at most $(8h^{-1} \log m)^m$ by (2.2) and the one-phase density bound, and their total averaging weight is $1/P$. Thus

$$\mathbb{E}\|Z - Z^{\text{id}}\|_2^2 \leq O(m^{-49}) + P^{-1}(8h^{-1} \log m)^m \longrightarrow 0,$$

since $P \geq m^{0.69m}$ and $h^{-1} = m^{0.60}$. Also

$$\|Z^{\text{id}} - 1\|_2^2 = A_*^m - 2(1 - 2m^{-2})^m + 1 \rightarrow 0$$

by (2.2) and $mh^2 = m^{-1/5} \rightarrow 0$. It follows that $\mathbb{E}\|Z - 1\|_2^2 \rightarrow 0$. There are only finitely many t at this stage. Markov's inequality, the two orientations, and the spread estimate therefore let us fix trial phases with strict spread inequalities and sampling norms below $1/(2j)$, simultaneously for every test. We can also avoid the finitely many equalities $n\lambda_i \in \mathbb{Z}$ and the endpoints $1/N, 2/N$, since each has probability zero.

Realizing the phases by circle characters. Now fix this m and these trial phases. Successively choose q_i as multiples of the preceding frequency so that $\{q_i\theta\}$ is as close to λ_i as desired and q_i/q_{i-1} is as large as desired. This is possible because the positive multiples of $q_{i-1}\theta$ are dense. More formally, make a sequence of such finite block choices for which all new ratios tend to infinity and all $\alpha_i \rightarrow \lambda_i$. The Haar law of $(u_i(x))_{i \in J_j}$ then converges weakly to product Haar measure. For each fixed nonzero integer character vector, the last nonzero coefficient times its frequency dominates the sum of the earlier terms, once the ratios are sufficiently large. That Fourier coefficient is therefore exactly zero eventually. Trigonometric approximation proves the weak convergence. The joint law including the deterministic phases converges to product Haar in the coordinates and the fixed vector of trial phases.

For fixed m the finitely many squared sampling tests are bounded functions of these coordinates and phases. Their discontinuities have zero limiting measure: every selected $\{n\lambda_i\}$ avoids the phase cut, and for that phase only finitely many coordinate values can lie on a cutoff or interval boundary. Weak convergence therefore gives convergence of their integrals. This familiar extension from continuous tests follows, for example, by continuous upper and lower approximations outside open neighborhoods of the null discontinuity sets. It applies to the squared differences from one as well. Choose a sufficiently late finite block choice. Its sampling norms are below $1/j$, all the strict spread and angle bounds persist, and all frequency ratios determined so far satisfy their required bounds. The ratio connecting the previous block to this one can be made large enough to meet that block's outstanding condition; the ratio at the end of this block is imposed at the next stage. This completes the induction. $\square$

Two consequences of the construction will be used repeatedly:

$$\sum_{s>j} \frac{1}{K_s} \leq \frac{2}{K_{j+1}}, \quad \sum_j \frac{1+h_j^*}{K_j} < \infty, \quad \frac{Q_j}{K_{j+1}} \rightarrow 0. \quad (2.14)$$

They follow respectively from $K_{j+1} > 2K_j$, the final inequality in (2.9), and $7/10 < 3/4$. The parameters are now fixed independently of any putative invariant projection. We next turn their sampling identities into operator identities between pairs of points of the circle.

3 Projection fields and transfer between endpoints

We now turn the parameters of Proposition 2.3 into a weighted shift. The purpose of this section is to convert invariance under that shift into an identity between projection fields at two different points of the circle. We first obtain the identity along positive rotation orbits, and then use the sampling estimates to reach every translation by an integer multiple of $1/q_A$, for some $A \geq 1$.

3.1 Fields determined by scalar multipliers

We first isolate the elementary field representation used here and in the direct weighted-shift construction. It depends only on commutation with scalar multipliers.

Lemma 3.1. *Let (Y, ν) be a probability space, let $K = \ell^2(\mathbb{Z})$, and let $A \in B(L^2(Y, \nu; K))$ commute with every scalar multiplication operator $M_g \zeta(y) = g(y)\zeta(y)$, $g \in L^\infty(Y, \nu)$. There is a field $A(y) \in B(K)$, unique almost everywhere, such that*

$$(A\zeta)(y) = A(y)\zeta(y), \quad \|A(y)\| \leq \|A\| \quad \text{almost everywhere.}$$

The field is measurable in the sense that $y \mapsto A(y)v$ is measurable for every $v \in K$. Products and adjoints of such operators are computed fiberwise. In particular, an orthogonal projection is represented by orthogonal projections outside one null set.

Proof. If $B \in B(L^2(Y, \nu))$ commutes with all multipliers, put $b = B1$. For every measurable set E ,

$$B\mathbf{1}_E = \mathbf{1}_E b, \quad \int_E |b|^2 d\nu \leq \|B\|^2 \nu(E).$$

Thus $|b| \leq \|B\|$ almost everywhere. The identity $Bf = bf$ follows first for simple functions and then, by density, for all $f \in L^2(Y)$. Apply this observation to the matrix entries of A between the coordinate copies of $L^2(Y)$, obtaining measurable functions a_{ij} .

Let v range over the countable set of finitely supported vectors with rational real and imaginary parts. Commutation with $M_{\mathbf{1}_E}$ and the operator bound give

$$\int_E \sum_{i \in \mathbb{Z}} \left| \sum_j a_{ij}(y) v_j \right|^2 d\nu(y) \leq \|A\|^2 \|v\|^2 \nu(E).$$

Consequently the integrand is at most $\|A\|^2 \|v\|^2$ almost everywhere. Outside the union of these countably many null sets, the matrix defines an operator $A(y)$ on K with norm at most $\|A\|$. Its columns, and then its values on every vector, are measurable by finite-coordinate approximation. The represented field agrees with A on finite-coordinate simple functions, hence on all of $L^2(Y; K)$.

Equality of two such fields follows by applying their represented operators to each constant basis field. Products act fiberwise by the representation formula, and the entries of the adjoint field are $\overline{a_{ji}}$. This proves the assertions about products and adjoints. The identities $p = p^* = p^2$ then give orthogonal projections on a common conull set. $\square$

3.2 The tracial representation

We use the standard crossed-product representation for a probability-preserving action; see Anantharaman and Popa [1, Section 1.4 and Proposition 1.4.5]. We verify below the field and trace properties needed in the proof.

Write $X = \mathbb{R}/\mathbb{Z}$, let μ be Haar probability measure, and put $\sigma x = x + \theta$. On $\mathcal{H} = L^2(X, \mu; \ell^2(\mathbb{Z}))$, let $Se_k = e_{k+1}$ and define

$$U(x) = S, \quad \pi(a)(x)e_k = a(\sigma^k x)e_k \quad (a \in L^\infty(X, \mu)).$$

For $V = \pi(x \mapsto e^{2\pi i x})$, these operators satisfy $VU = e^{2\pi i \theta} UV$. We also write e_0 for the constant field with this value. The vector $U^m V^n e_0$ is the field $x \mapsto e^{2\pi i n x} e_m$. These vectors span a dense subspace by ordinary Fourier theory, and $\langle U^m V^n e_0, e_0 \rangle$ equals 1 for $(m, n) = (0, 0)$ and 0 otherwise. The commutation relation reduces every product of monomials to a scalar

multiple of a monomial, so these moments identify all inner products of polynomial vectors with the prescribed tracial cyclic representation. The resulting unitary identifies the generated algebra M with R_θ and its faithful normal trace with τ . Measurable functional calculus for V places every $\pi(a)$ in M .

Every generator commutes with scalar multiplication by a function of x . Lemma 3.1 therefore represents every $A \in M$ by an essentially bounded measurable field. The generators also commute with the unitary J given by $(J\zeta)(x) = S\zeta(\sigma x)$: for $\pi(a)$ the coordinate calculation is $a(\sigma^{k-1}(\sigma x)) = a(\sigma^k x)$. These commutations pass to their von Neumann algebra. Testing $AJ = JA$ on the constant basis fields gives

$$A(\sigma x) = S^{-1}A(x)S \quad \text{for almost every } x. \quad (3.1)$$

The vector formula for the trace is now

$$\tau(A) = \int_X \langle A(x)e_0, e_0 \rangle d\mu(x) \quad (A \in M). \quad (3.2)$$

We choose Borel versions for projection fields and discard null sets and their countably many rotation translates when necessary. Irrational rotation is ergodic: all Fourier coefficients of nonzero index vanish for an invariant L^2 function. We use equidistribution in the form established at the beginning of Section 2.

The following consequence of covariance will also apply to the product model. It ensures that a nontrivial invariant projection has a nonzero zeroth coordinate at almost every endpoint, as does its complement.

Lemma 3.2. *Let σ be an invertible ergodic probability-preserving transformation of (Y, ν) . On $L^2(Y, \nu; \ell^2(\mathbb{Z}))$, define $(U\zeta)(y) = S\zeta(y)$ and $\pi(a)(y)e_k = a(\sigma^k y)e_k$, where $Se_k = e_{k+1}$. Suppose a von Neumann algebra M containing U and $\pi(L^\infty(Y))$ consists of measurable fields with covariance $A(\sigma y) = S^{-1}A(y)S$ and has faithful normal tracial state*

$$\tau(A) = \int_Y \langle A(y)e_0, e_0 \rangle d\nu(y).$$

Let $f \in L^\infty(Y, \nu)$ be positive almost everywhere, put $T = U\pi(f)$, and let $p \in M$ be an orthogonal projection with $(1 - p)Tp = 0$. If $0 < \tau(p) < 1$, then $0 < \langle p(y)e_0, e_0 \rangle < 1$ almost everywhere.

Proof. Put $r(y) = \langle p(y)e_0, e_0 \rangle$ and $D_y = \text{diag}_k f(\sigma^k y)$. Invariance and covariance give $D_y \text{ran } p(y) \subseteq \text{ran } p(\sigma y)$. If $r(\sigma y) = 0$, every vector in the latter range has zero zeroth coordinate. Since $(D_y v)_0 = f(y)v_0$ and $f(y) > 0$, every vector in the former range also has zero zeroth coordinate, so $r(y) = 0$. If $r(y) = 1$, then e_0 belongs to the former range, and its image $f(y)e_0$ shows that $r(\sigma y) = 1$.

Thus $\sigma^{-1}\{r = 0\} \subseteq \{r = 0\}$ and $\{r = 1\} \subseteq \sigma^{-1}\{r = 1\}$ modulo null sets. Probability preservation makes both inclusions equalities modulo null sets. Ergodicity makes each event null or conull. The trace formula and $0 < \tau(p) < 1$ exclude both conull cases. $\square$

3.3 A compact-set transfer lemma

Projection fields need not be continuous. The following observation allows us to pass an identity to a weak limit of endpoint measures, provided that their marginals do not concentrate on small sets. A family of finite measures is *uniformly absolutely continuous* with respect to μ if, for each $\varepsilon > 0$, one can choose $\delta > 0$ such that every set of μ -measure below δ has measure below ε for every member of the family.

Lemma 3.3. *Let X be a compact metric space with Borel probability measure μ . Let $p(x)$ be a Borel field of orthogonal projections on $\ell^2(\mathbb{Z})$, meaning that every column $x \mapsto p(x)e_k$ is Borel, and let $F_k : X \rightarrow (0, \infty)$, $k \in \mathbb{Z}$, be finite Borel functions. Suppose finite positive Borel measures β_l on X^2 converge weakly to β . Assume that both families of marginals of β_l are uniformly absolutely continuous with respect to μ , and that both marginals of β are absolutely continuous. Suppose also that some $0 < B < \infty$ satisfies*

$$F_k(y)/F_k(x) \leq B \quad (k \in \mathbb{Z})$$

almost everywhere for each β_l and for β . If

$$(1 - p(y)) \operatorname{diag}_k(F_k(y)/F_k(x))p(x) = 0 \quad (3.3)$$

holds β_l -almost everywhere for every l , then it holds β -almost everywhere.

Proof. First replace every diagonal entry by its minimum with B , so that the diagonal field is bounded everywhere. Given an arbitrarily small $\varepsilon > 0$, simultaneous Lusin approximation supplies a compact $Q \subset X$ on which every F_k and every vector map $x \mapsto p(x)e_k$ is continuous, while $\mu(X \setminus Q)$ is sufficiently small to make the β_l -mass outside Q^2 less than ε for all l . For completeness, the simultaneous assertion follows by applying Lusin's property to the countably many maps with summable exceptional measures and intersecting the resulting compact sets. The usual property for maps into separable metric spaces follows by finite-valued approximation and measure regularity.

On Q^2 , the capped diagonal field is strongly continuous: continuity on the basis vectors and its uniform bound imply continuity on every vector. The same argument applies to $p(x)$ and $p(y)$. Products of strongly continuous, uniformly bounded fields are strongly continuous. Therefore the zero set Z_Q of the capped version of (3.3) in Q^2 is compact. The assumed identity implies $\beta_l(Z_Q) \geq \beta_l(X^2) - \varepsilon$. The compact-set upper bound under weak convergence and convergence of total masses give

$$\beta(Z_Q) \geq \limsup_l \beta_l(Z_Q) \geq \beta(X^2) - \varepsilon.$$

The capped global zero set thus has full β -measure, since ε was arbitrary. It is measurable by its countably many matrix entries. Finally the cap changes no diagonal entry β -almost everywhere. $\square$

3.4 The measurable weight and its cocycles

We return to the prescribed rotation on $X = \mathbb{R}/\mathbb{Z}$. The next formulas supply its positive weight and the diagonal ratios to which Lemma 3.3 will apply. For the parameters of Proposition 2.3, put

$$g_*(x) = \sum_j C_j \sum_{i \in J_j} [u_i(x) + \alpha_i], \quad f_*(x) = e^{-g_*(x)}, \quad T_* = U\pi(f_*). \quad (3.4)$$

The sum defining g_* has only finitely many nonzero terms almost everywhere. Here $c_i = C_j$ for $i \in J_j$ in the introduction's notation. To prove the finiteness assertion, observe that

$$\sum_i \mu\{[u_i + \alpha_i] \neq 0\} = \sum_i \alpha_i \leq 2 \sum_j K_j^{-1} < \infty.$$

The elementary first Borel–Cantelli implication follows here simply by bounding the measure of a union of tail events by the sum of their measures. Thus $0 < f_* \leq 1$ almost everywhere.

For each $k \in \mathbb{Z}$, define

$$F_k(x) = \exp \left(- \sum_j C_j \sum_{i \in J_j} [u_i(x) + k\alpha_i] \right). \quad (3.5)$$

These sums, too, have only finitely many nonzero terms almost everywhere: in the tail where $|k|\alpha_i < 1$, the probability of a nonzero floor is $|k|\alpha_i$. After taking a countable intersection, there is an invariant conull set on which all F_k are finite and strictly positive and

$$F_{v+n}(x) = F_v(x)F_n(\sigma^v x), \quad F_0 = 1, \quad F_1 = f_*. \quad (3.6)$$

Indeed $u_i(\sigma^v x) = \{u_i(x) + v\alpha_i\}$, so the addition formula for floors proves the identity term by term. Since $\alpha_i > 0$, $F_k(x)$ is nonincreasing in k . Give these functions arbitrary positive finite Borel values on the remaining null set.

For endpoints in the common conull set, write

$$R_{x,y}(k) = F_k(y)/F_k(x).$$

If $u_i(x) \leq u_i(y)$, the i th coordinate contributes at most 1 to this ratio. For $i \in J_s$, it contributes e^{-C_s} precisely when

$$u_i(x) < 1 - \{k\alpha_i\} \leq u_i(y). \quad (3.7)$$

When $k\alpha_i$ is integral the displayed event is empty, as it should be. Without an order assumption, an individual coordinate contributes at most e^{C_s} , since the two floors differ by at most one.

3.5 From invariant projections to endpoint identities

Suppose $p \in M$ is a projection satisfying $(1 - p)T_*p = 0$. Its range is then invariant under every positive power of T_* . The formula

$$T_*(x)^n = S^n \operatorname{diag}_v F_n(\sigma^v x)$$

and covariance (3.1) show that, for $y = \sigma^n x$,

$$(1 - p(y)) \operatorname{diag}_v R_{x,y}(v)p(x) = 0 \quad (n \geq 1). \quad (3.8)$$

To verify the division involved, the cocycle identity gives $F_n(\sigma^v x)/F_n(x) = F_v(\sigma^n x)/F_v(x)$; the positive divisor $F_n(x)$ is finite. The resulting diagonal is bounded on each such fiber by $1/F_n(x)$, since $F_n \leq 1$. This bound need not be uniform in x .

If $0 < \tau(p) < 1$, then

$$0 < r(x) := \langle p(x)e_0, e_0 \rangle < 1 \quad \text{for almost every } x. \quad (3.9)$$

This is Lemma 3.2, applied with $f = f_*$, the faithful trace (3.2), and covariance (3.1).

The next lemma is the payoff from the sampling construction. It extends (3.8) beyond actual orbit pairs, while retaining a bound on every diagonal entry at once.

Lemma 3.4. *Let $p \in M$ be a projection satisfying $(1 - p)T_*p = 0$. For every integer $A \geq 1$ and every $t \in X$ that is an integer multiple of $1/q_A$, the identity*

$$(1 - p(x + t)) \operatorname{diag}_v R_{x,x+t}(v)p(x) = 0 \quad (3.10)$$

holds for almost every x . The exceptional sets may be chosen simultaneously for this countable family of translations.

Proof. Fix A, t . In every sufficiently late stage j , put $a = d_{j-1} \geq A$. Since q_A divides q_a , the sampling set $\mathcal{N}_{j,t}$ occurs in Proposition 2.3. Write $w = w_{m_j}$ and $w^{(2)} = w_{m_j}^{(2)}$ for this stage's kernels in (2.1). Average over $n \in \mathcal{N}_{j,t}$ the graph measures $x \mapsto (x, \sigma^n x)$ with starting density

$$\prod_{i \in J_j} w(u_i(x), \{n\alpha_i\}).$$

By (2.11), both marginal densities tend to 1 in $L^2(\mu)$. For the second marginal this uses the second orientation $w^{(2)}(z, v) = w(z - v, v)$. As every translation $n\theta$ lies within ℓ_j of t and $\ell_j \rightarrow 0$, testing continuous functions on X^2 shows that these measures converge weakly to the Haar-first graph measure of $x \mapsto x + t$. At every point of positive density,

$$0 < \{n\alpha_i\} < m_j^{-3/5} \Delta(u_i(x)) \quad (i \in J_j),$$

so there is no wrap and all current-block coordinates are ordered.

For every selected n , remove the same set of starting points, namely those satisfying at least one of

$$\begin{aligned} u_i(x) &\geq 1 - q_a \ell_j & (A \leq i \leq a), \\ u_i(x) &\geq 1 - 2Q_j/N_s & (i \in J_s, \ s > j). \end{aligned}$$

The removed Haar measure is at most

$$aq_a \ell_j + 2Q_j \sum_{s > j} K_s^{-1} \leq (j+1)^{-2} + 4Q_j/K_{j+1} \rightarrow 0.$$

The L^2 bound on the averaged first marginal and Cauchy–Schwarz show that the removed weighted mass tends to zero. Thus the trimmed measures have the same weak limit. Each trimmed marginal is dominated by its untrimmed counterpart, so both retain uniform L^2 bounds and hence uniform absolute continuity.

For $A \leq i \leq a$, the number $q_i t$ is integral. The coordinate advance is therefore $q_i \{n\theta - t\} < q_a \ell_j$, and trimming prevents a wrap. For future coordinates it is $n\alpha_i < 2Q_j/N_s$, again without a wrap. Together with the current block, this orders every coordinate $i \geq A$. Hence, on all trimmed measures,

$$R_{x,y}(v) \leq \exp \left(\sum_{i < A} C_{s(i)} \right) \quad (v \in \mathbb{Z}), \quad i \in J_{s(i)}.$$

The bound is independent of j and v . It also holds on the limiting graph, because $u_i(x+t) = u_i(x)$ for $i \geq A$. The positive-orbit identity holds on the approximants, so Lemma 3.3 gives (3.10). There are only countably many pairs (A, t) under consideration. $\square$

4 Ordered pairs from independent digits

The rational-translation identities of Lemma 3.4 will now be transferred to pairs with many opportunities to cross the thresholds in (3.7). If both endpoint laws were Haar, coordinate order would make each nonnegative difference $u_i(y) - u_i(x)$ have mean zero, leaving no threshold crossings. We therefore allow a controlled change in the endpoint marginals. What matters is a uniform square-integrable bound for their marginal densities and a product structure that survives conditioning on either endpoint. We first isolate the density estimate, which also applies to the independent-coordinate construction in the appendix.

Lemma 4.1 (Densities of independent coordinates). *Let $(\Omega_i, \mathcal{F}_i, \mu_i)$, $i \geq 1$, be probability spaces and let $h_i \geq 0$ satisfy $\int h_i d\mu_i = 1$. If*

$$\sum_{i \geq 1} \|h_i - 1\|_{L^2(\mu_i)}^2 \leq C < \infty,$$

then the product probability measure $\nu = \bigotimes_{i \geq 1} (h_i \mu_i)$ has a density with respect to $\mu = \bigotimes_{i \geq 1} \mu_i$ whose $L^2(\mu)$ norm is at most $e^{C/2}$.

In particular, suppose two probability densities h_i^0, h_i^1 at each coordinate satisfy

$$\|h_i^a - 1\|_{L^2(\mu_i)}^2 \leq b_i \quad (a = 0, 1), \quad \sum_{i \geq 1} b_i \leq C.$$

For every deterministic sequence $\epsilon_i \in \{0, 1\}$, the product law with coordinate densities $h_i^{\epsilon_i}$ has the same bound.

Proof. The mean-one assumption gives $\|h_i\|_2^2 = 1 + \|h_i - 1\|_2^2$. Hence each finite product density obeys

$$\left\| \prod_{i=1}^n h_i \right\|_{L^2(\mu)}^2 = \prod_{i=1}^n (1 + \|h_i - 1\|_2^2) \leq \exp \left(\sum_{i=1}^n \|h_i - 1\|_2^2 \right) \leq e^C.$$

For a bounded cylinder function φ depending on the first n coordinates, integration under ν is therefore bounded in absolute value by $e^{C/2} \|\varphi\|_{L^2(\mu)}$. These cylinder functions are dense in $L^2(\mu)$, so the integration functional extends continuously to that space. The Riesz representation theorem gives a real $H \in L^2(\mu)$ with norm at most $e^{C/2}$ representing the extension. The finite signed measure $H\mu$ and the probability measure ν agree on all cylinder sets. Uniqueness of finite measures on the product sigma-algebra gives $H\mu = \nu$; thus $H \geq 0$ almost everywhere and $\int H d\mu = 1$. Applying the first assertion to $h_i = h_i^{\epsilon_i}$ proves the hybrid assertion. $\square$

Write $r_i^* = q_{i+1}/q_i$. A Haar-uniform point of X has the expansion

$$x = \frac{A_0}{q_1} + \sum_{i \geq 1} \frac{A_i}{q_{i+1}}, \quad A_0 \in \{0, \dots, q_1 - 1\}, \quad A_i \in \{0, \dots, r_i^* - 1\}, \quad (4.1)$$

with independent uniform digits. Indeed, specifying a finite prefix selects an interval whose length is the reciprocal of the product of the relevant radices. Conversely these intervals determine the Haar measure. The ambiguity at endpoints of such intervals is a null set. In particular, $A_i = \lfloor r_i^* u_i(x) \rfloor$, and $u_i(x)$ is the tail of (4.1) starting with digit A_i , after multiplication by q_i . We couple the independent digits; each character coordinate u_i depends on an overlapping tail of these digits.

Fix a block index j . We construct a probability law $\mathbb{P}_j$ on a pair (x, y) , retaining auxiliary labels. Use the same uniform initial digit for both endpoints. Most coordinate pairs will have equal raw values. Rare top and bottom components force crossings even after one endpoint has been fixed, and an extra component in block j places the endpoints close to opposite ends of the unit interval. These components give tail attenuation while keeping the marginal densities uniformly square integrable. Independently at every $i \in J_s$, choose a raw pair (X_i, Y_i) by the following mixture, where

$$\rho_s(w) = \kappa_s \sqrt{\frac{w}{m_s}}.$$

For each of the two labelled widths $w = a_s, b_s$, include these two components, each of probability $\rho_s(w)$:

$$\begin{array}{ll} \text{top:} & X_i \text{ uniform on } [0, 1 - w/2), & Y_i \text{ uniform on } [1 - w/2, 1), \\ \text{bottom:} & X_i \text{ uniform on } [0, w/2), & Y_i \text{ uniform on } [w/2, 1). \end{array}$$

The two choices within each displayed component are independent. Keep the width labels distinct even if their numerical values coincide. If $s = j$, also include a *tagged* component of probability $1/m_s$, with independent uniforms

$$X_i \in [0, 1/(2m_s)), \quad Y_i \in [1 - 1/(2m_s), 1).$$

With the remaining probability set $X_i = Y_i$ to a uniform point of $[0, 1)$. These probabilities have a positive remainder: the parameter bounds give $\rho_s(w)/w \leq \kappa_s$, $w \leq 1/4$, and $m_s \geq 200$. Define the endpoint digits by

$$A_i = \lfloor r_i^* X_i \rfloor, \quad B_i = \lfloor r_i^* Y_i \rfloor,$$

and use (4.1) to form x from the A_i and y from the B_i . Thus X_i, Y_i choose the first digit of the corresponding circle coordinates; they are not themselves those coordinates.

Lemma 4.2 (Uniform marginal bounds). *There is a finite constant H_0 , independent of j , such that the marginal laws of x and y have densities with respect to μ of $L^2(\mu)$ norm at most H_0 . The same assertion holds for an endpoint formed by any deterministic choice of the x -digit or the y -digit at each coordinate, retaining the common uniform initial digit.*

Proof. Either raw marginal density is at most 10. For each width its top and bottom contributions together are at most $4\rho_s(w)/w$, the tag contributes at most 2, and the diagonal component contributes at most 1.

We also need a sharper integral estimate. A uniform density on an interval of length at least $w/2$ differs from 1 by a mean-zero function of squared L^2 norm at most $2/w$. For a tagged interval the corresponding bound is $2m_s$. Express a raw marginal density minus 1 as the sum of its at most five weighted mean-zero perturbations. The finite-sum Cauchy–Schwarz inequality gives

$$\int_0^1 d_i(t)^2 dt \leq 1 + 50 \left(\frac{\kappa_s^2}{m_s} + \frac{\mathbf{1}_{\{s=j\}}}{m_s} \right), \quad i \in J_s. \quad (4.2)$$

In fact the four width terms contribute at most $40\kappa_s^2/m_s$, and the tag at most $10/m_s$. Passing from raw variables to digits replaces d_i by its average on each radix bin, so Jensen’s inequality preserves (4.2).

Use the uniform law at each digit as the reference probability measure. The pairs of digits are independent across coordinates, and each endpoint digit density has mean one. Since $|J_s| = m_s$, summing (4.2) minus one over the coordinates gives the common bound $50(\sum_s \kappa_s^2 + 1)$, for either endpoint and every deterministic choice between them. The initial digit has density one. Lemma 4.1 therefore gives the bound

$$H_0^2 := \exp \left(50 \left(\sum_{s \geq 1} \kappa_s^2 + 1 \right) \right) < \infty.$$

Outside the null sets of ambiguous expansions, the mixed-radix map is a measure-preserving bijection from the reference digit space to X . Transporting the density through this map gives a Haar density with the same L^2 norm. This proves the assertion for both endpoints and all the deterministic hybrids. $\square$

All the endpoint expansions consequently avoid their ambiguous values almost surely. Every raw pair satisfies $X_i \leq Y_i$, so $A_i \leq B_i$. The nonnegative tail expansion then gives

$$u_i(x) \leq u_i(y), \quad |u_i(x) - X_i| \leq \frac{1}{r_i^*}, \quad |u_i(y) - Y_i| \leq \frac{1}{r_i^*} \quad (i \geq 1). \quad (4.3)$$

For $D \geq 1$, let $y^{(D)}$ have y 's digits at indices $1, \dots, D$ and x 's later digits. Then $u_i(x) \leq u_i(y^{(D)})$ for all i , $y^{(D)} \rightarrow y$, and $y^{(D)} - x$ is a multiple of $1/q_{D+1}$.

Proposition 4.3 (Invariance on the ordered pairs). *Let $p \in M$ be a projection satisfying $(1 - p)T_*p = 0$. For every j ,*

$$(1 - p(y)) \operatorname{diag}_{k \in \mathbb{Z}}(R_{x,y}(k))p(x) = 0 \quad \text{for } \mathbb{P}_j\text{-almost every pair.} \quad (4.4)$$

Moreover $0 < R_{x,y}(k) \leq 1$ for all $k \in \mathbb{Z}$ almost surely.

Proof. Fix D . The rational shifts $y^{(D)} - x$ have only finitely many possible values. Lemma 3.4 holds outside a common Haar-null set for all these shifts, so the absolute continuity of the x marginal implies the required identity at $(x, y^{(D)})$. This remains valid even though the selected shift depends on the pair.

The coordinate order and (3.7) bound every ratio by 1, both for these pairs and for their limit (x, y) . Their measures converge weakly by the almost-sure convergence $y^{(D)} \rightarrow y$. Lemma 4.2 supplies uniform L^2 bounds for both marginals. In particular their masses on a set of Haar measure δ are at most $H_0\sqrt{\delta}$. Lemma 3.3 now passes the identity to the limit. $\square$

We have transferred the invariance identity to ordered pairs. The remaining reason for choosing these particular laws is that an interior threshold has a quantitatively positive chance of being crossed, even after one entire endpoint has been fixed.

Lemma 4.4 (Conditional labels and forced crossings). *Conditional on x , all coordinate labels are independent and each posterior depends only on the corresponding digit A_i . The analogous statement holds conditional on y , using B_i .*

For $i \in J_s$, $w \in \{a_s, b_s\}$, and $u \in [w, 1 - w]$, the following bounds hold almost surely:

$$\mathbb{P}_j(\text{top label of width } w \text{ at } i \mid x) \geq \rho_s(w)/10 \quad \text{if } u_i(x) < u, \quad (4.5)$$

$$\mathbb{P}_j(\text{bottom label of width } w \text{ at } i \mid y) \geq \rho_s(w)/10 \quad \text{if } u_i(y) \geq u. \quad (4.6)$$

In the first case that label forces $u_i(y) \geq u$; in the second it forces $u_i(x) < u$.

Proof. The entire endpoint determines all its digits almost surely. For any finite collection of labels, conditioning on all x -digits gives the product of the individual posteriors

$$\frac{\mathbb{P}_j(\text{label } \ell \text{ at } i) \mathbb{P}_j(A_i = a \mid \text{label } \ell \text{ at } i)}{\mathbb{P}_j(A_i = a)}.$$

One verifies this formula first against cylinder tests in the digit sequence, using independence of coordinate pairs, and then against all bounded digit-measurable tests by the monotone class theorem. The common initial digit is independent of those pairs. This proves conditional independence given x ; the proof for y is the same. In particular, the dependence of $u_i(x)$ on later digits does not change the label posteriors after the entire endpoint has been conditioned on.

The parameters satisfy

$$\frac{1}{r_i^*} \leq \frac{1}{N_s} < \frac{w}{2}.$$

If $u_i(x) < u \leq 1 - w$, the whole radix bin containing X_i lies in $[0, 1 - w/2)$. On that bin the top component's X_i -density is at least 1, while the total raw marginal density is at most 10. Its posterior probability is therefore at least $\rho_s(w)/10$. Under this label, (4.3) gives

$$u_i(y) \geq 1 - w/2 - 1/r_i^* > 1 - w \geq u.$$

For the second assertion, $u_i(y) \geq u \geq w$ places the entire Y_i bin in $[w/2, 1)$. The bottom component again has density at least 1 there, so its posterior is at least $\rho_s(w)/10$. Its X_i value satisfies

$$u_i(x) \leq w/2 + 1/r_i^* < w \leq u.$$

These estimates include thresholds at radix-bin endpoints. $\square$

5 The convolution obstruction

We can now exclude every nontrivial invariant projection for the measurable weight. The argument compares two estimates for a finite part of a convolution: polynomial compactness prevents that part from vanishing, whereas the ordered couplings make its complementary tail too small to cancel it.

5.1 A conditional convolution estimate

We isolate the common argument so that the product construction can use it without repeating the infinite-tail estimate. The probability space in the lemma may include auxiliary labels besides its endpoints.

Lemma 5.1 (Conditional convolution obstruction). *Let $K \geq 1$ be an integer, $D \geq a > 0$, $c > 0$, and $\eta, \delta \geq 0$. On a probability space with endpoints x, y , let $\xi(x), \gamma(y)$ be measurable vectors in the unit ball of $\ell^2(\mathbb{Z})$, and let F_k be positive finite measurable functions of an endpoint, with $F_0 = 1$. Set*

$$A_v = \frac{\xi_v(x)}{F_v(x)}, \quad B_w = F_{-w}(y) \overline{\gamma_{-w}(y)}, \quad R(k) = \frac{F_k(y)}{F_k(x)}.$$

Suppose, almost surely, for $0 \leq n \leq 2K$ and every $v \in \mathbb{Z}$,

$$\sum_{v \in \mathbb{Z}} A_v B_{-n-v} = 0, \quad 0 < \frac{F_{n+v}(y)}{F_v(x)} \leq \min\{R(v), R(n+v)\} \leq 1. \quad (5.1)$$

Let E be an event of probability at least c on which

$$|A_v|, |B_v| \leq D \quad (|v| \leq K), \quad |A_0|, |B_0| \geq a.$$

For each $|k| > K$, suppose there are events $H_k^{(1)}, H_k^{(2)}$ with

$$R(k) \leq \eta + \mathbf{1}_{H_k^{(1)}} + \mathbf{1}_{H_k^{(2)}} \quad \text{on } E, \quad \mathbb{P}(H_k^{(1)} \mid x) \leq \delta, \quad \mathbb{P}(H_k^{(2)} \mid y) \leq \delta. \quad (5.2)$$

The conditional bounds are under the full probability law, before restriction to E . If $\varepsilon(K, D, a)$ is the positive constant from Lemma 2.2, then

$$c\varepsilon(K, D, a) \leq (2K + 1)(\eta + 2\sqrt{\delta}). \quad (5.3)$$

Proof. The multiplier bound in (5.1) and Cauchy–Schwarz make every convolution there absolutely convergent. Put

$$c_n = \sum_{\substack{|v| \leq K \\ |n+v| \leq K}} A_v B_{-n-v} \quad (0 \leq n \leq 2K).$$

On E , Lemma 2.2 applied to the two truncated Laurent polynomials gives $\sum_{n=0}^{2K} |c_n| \geq \varepsilon(K, D, a)$. Its expected value on E is therefore at least $c\varepsilon(K, D, a)$.

Fix n . For an omitted index v , choose $k(v) = v$ if $|v| > K$, and $k(v) = n + v$ otherwise. This is a deterministic choice with $|k(v)| > K$. The zero convolution and the two multiplier bounds give

$$\mathbf{1}_E |c_n| \leq \eta + \sum_{b=1}^2 \sum_{v \text{ omitted}} |\xi_v(x) \gamma_{n+v}(y)| \mathbf{1}_{H_{k(v)}^{(b)}}. \quad (5.4)$$

The unmasked sum of products of vector coordinates is at most one, which accounts for η . We dropped $\mathbf{1}_E$ from the nonnegative exceptional terms so that their full-law conditional bounds remain applicable.

For the first exceptional sum, Cauchy–Schwarz on probability times counting measure bounds its expectation by

$$\left(\mathbb{E} \sum_{v \text{ omitted}} |\xi_v(x)|^2 \mathbf{1}_{H_{k(v)}^{(1)}} \right)^{1/2} \left(\mathbb{E} \sum_{v \text{ omitted}} |\gamma_{n+v}(y)|^2 \right)^{1/2} \leq \sqrt{\delta}.$$

Indeed, condition the first factor's square on x , apply (5.2) to each deterministic $k(v)$, and use $\sum_v |\xi_v(x)|^2 \leq 1$. The second factor is at most one. For the other exceptional sum put its indicator with $|\gamma_{n+v}(y)|^2$ and condition on y , giving the same bound. Tonelli's theorem justifies every interchange of the nonnegative sums and integrals. No sum of the individual exceptional probabilities is required. Taking expectations in (5.4) and summing in n proves (5.3). $\square$

5.2 Application to the circle weight

Theorem 5.2. *The bounded operator $T_* = U\pi(f_*)$ has no projection $p \in M$ satisfying $0 < \tau(p) < 1$ and $(1 - p)T_*p = 0$.*

Suppose, for contradiction, that such a projection exists. Put

$$\xi(x) = p(x)e_0, \quad \gamma(y) = (1 - p(y))e_0, \quad r(x) = p(x)e_{0,0}.$$

Both vector norms are at most 1. We shall use the laws $\mathbb{P}_j$ of Section 4; their expectations are denoted by $\mathbb{E}_j$.

5.3 The exact convolution identities

By (4.4), the vector $\text{diag}_v(R_{x,y}(v))\xi(x)$ belongs to $\text{ran } p(y)$. Applying $T_*(y)^n$, which preserves that range for $n \geq 0$, and pairing with $\gamma(y)$ gives

$$0 = \sum_{v \in \mathbb{Z}} \frac{F_{n+v}(y)}{F_v(x)} \xi_v(x) \overline{\gamma_{n+v}(y)} = \sum_{v \in \mathbb{Z}} A_v B_{-n-v}, \quad n \geq 0, \quad (5.5)$$

where, in this section,

$$A_v = \frac{\xi_v(x)}{F_v(x)}, \quad B_v = F_{-v}(y) \overline{\gamma_{-v}(y)}.$$

All these identities hold simultaneously almost surely under each $\mathbb{P}_j$. They are absolutely convergent. Indeed, monotonicity of F_k in k and coordinate order imply

$$\frac{F_{n+v}(y)}{F_v(x)} \leq \min\{R_{x,y}(v), R_{x,y}(n+v)\} \leq 1, \quad (5.6)$$

and $\sum_v |\xi_v(x) \gamma_{n+v}(y)| \leq 1$ by Cauchy–Schwarz. The comparison with $R_{x,y}(v)$ follows from $F_{n+v}(y) \leq F_v(y)$; the comparison with $R_{x,y}(n+v)$ follows from $F_{n+v}(x) \leq F_v(x)$.

5.4 A finite convolution cannot be too small

Fix j and abbreviate $K = K_j$, $m = m_j$. Define an event E_j by the following three conditions:

$$\begin{aligned} X_i &< 1 - 3/m, \quad Y_i > 3/m \quad (i \in J_j), \quad \text{and at least one tag in } J_j; \\ X_i &< 1 - 3K/N_s, \quad Y_i > 3K/N_s \quad (i \in J_s, \ s > j); \\ r(x) &\geq 1/j, \quad 1 - r(y) \geq 1/j. \end{aligned}$$

For $0 \leq n \leq 2K$, let

$$c_{j,n} = \sum_{\substack{|v| \leq K \\ |n+v| \leq K}} A_v B_{-n-v}. \quad (5.7)$$

Lemma 5.3. *There is a constant $c_* > 0$ such that, for all sufficiently large j ,*

$$\mathbb{P}_j(E_j) \geq c_*, \quad \mathbb{E}_j \left[\mathbf{1}_{E_j} \sum_{n=0}^{2K_j} |c_{j,n}| \right] \geq c_* \varepsilon_j. \quad (5.8)$$

The point after which these inequalities hold may depend on p .

Proof. At a current-block coordinate, a tag has probability $1/m$ and satisfies both endpoint exclusions. Since each raw marginal density is at most 10, the probability of an inadmissible outcome is at most $60/m$. Thus a nontagged admissible outcome has probability at least $1 - 61/m$. Independence across the block shows that exactly one tag and admissible nontagged outcomes at the other coordinates have probability at least

$$m \frac{1}{m} (1 - 61/m)^{m-1} \geq e^{-90} \quad (m \geq 200).$$

For example, the last estimate follows from $\log(1 - t) \geq -t/(1 - t)$ with $t = 61/m$.

The total probability excluded by the future-block inequalities is at most

$$60K_j \sum_{s>j} \frac{1}{K_s} \leq \frac{120K_j}{K_{j+1}} \rightarrow 0.$$

By (3.9), $0 < r < 1$ Haar-almost everywhere. Lemma 4.2 therefore bounds the probability lost from the last line of the definition of E_j by

$$H_0 \mu\{r < 1/j\}^{1/2} + H_0 \mu\{1 - r < 1/j\}^{1/2} \rightarrow 0.$$

Consequently $c_* = e^{-90}/2$ works for all sufficiently large j .

On E_j , rounding changes the raw endpoint inequalities by at most $1/N_s$. Since $K \geq 1$, they imply

$$u_i(x) < 1 - 2K/N_s, \quad u_i(y) > 2K/N_s \quad (i \in J_s, \ s \geq j).$$

For $0 \leq v \leq K$, the bound $\alpha_i < 2/N_s$ makes the floors of $u_i(x) + v\alpha_i$ and $u_i(y) - v\alpha_i$ zero in every such block. Earlier-block floors have absolute value at most $K + 1$. Hence

$$1/F_v(x) \leq D_j, \quad F_{-v}(y) \leq D_j \quad (0 \leq v \leq K).$$

For $v < 0$ these two quantities are at most 1, by monotonicity and $F_0 = 1$. Since the vector coordinates have modulus at most 1, all coefficients of the Laurent polynomials

$$A^{[K]}(z) = \sum_{|v| \leq K} A_v z^v, \quad B^{[K]}(z) = \sum_{|v| \leq K} B_v z^v$$

have modulus at most D_j . Their constant coefficients are respectively $r(x)$ and $1 - r(y)$, both at least $1/j$ on E_j . The coefficient of z^{-n} in their product is $c_{j,n}$. The polynomial bound (2.10) now gives $\sum_{n=0}^{2K} |c_{j,n}| \geq \varepsilon_j$ on E_j . Integration proves the second assertion. $\square$

5.5 Controlling every part of the infinite tail

To contradict (5.8), we must bound the omitted terms of (5.5). The important estimate is conditional on one endpoint. It will let us sum infinitely many terms without taking a union bound over their exceptional events.

Lemma 5.4. *For every j and every integer k with $|k| > K_j$, there are measurable events $H_k^{(1)}, H_k^{(2)}$ in the labelled coupling space such that*

$$\mathbb{P}_j(H_k^{(1)} \mid x) \leq e^{-L_j/400}, \quad \mathbb{P}_j(H_k^{(2)} \mid y) \leq e^{-L_j/400} \quad \text{almost surely,} \quad (5.9)$$

and on E_j ,

$$R_{x,y}(k) \leq e^{-L_j} + \mathbf{1}_{H_k^{(1)}} + \mathbf{1}_{H_k^{(2)}}. \quad (5.10)$$

The conditional bounds hold before any restriction to E_j .

Proof. First suppose $K_j < |k| \leq N_j/4$. A tagged coordinate in J_j satisfies, after rounding,

$$u_i(x) < 1/m_j, \quad u_i(y) > 1 - 1/m_j.$$

Here $\beta = |k|\alpha_i$ lies in $(1/m_j, 1/2)$. The threshold $1 - \{k\alpha_i\}$ equals $1 - \beta$ for $k > 0$ and β for $k < 0$. It is crossed in either case, so (3.7) gives $R_{x,y}(k) \leq e^{-C_j} \leq e^{-L_j}$ on E_j . Set both exceptional events to be empty in this range.

We next cover all larger $|k|$. Put $M_s = m_s^{(4/5)m_s}$. The intervals

$$[N_j/4, M_j], \quad [G_s K_s, M_s] \quad (s > j)$$

cover $[N_j/4, \infty)$. Indeed the next left endpoint is below the preceding right endpoint by (2.9), the right endpoints tend to infinity, and $G_s K_s \leq N_s/4$ because $m_s \geq 4G_s$. Split each later interval at $N_s/4$. In its lower part the thresholds belong to $[a_s, 1 - a_s]$ at every coordinate of J_s . In its upper part, (2.8) places them in $[b_s, 1 - b_s]$ at at least $m_s/20$ coordinates. The initial interval uses the upper-part assertion for $s = j$.

Thus for each fixed k in this tail we can choose, deterministically, a block J_s , a width $w \in \{a_s, b_s\}$, and a set $I \subset J_s$ of at least $m_s/20$ indices such that

$$u_i^* := 1 - \{k\alpha_i\} \in [w, 1 - w] \quad (i \in I).$$

The lower-part choice always has $s > j$. In either type of choice,

$$m_s \rho_s(w) \geq L_j, \quad C_s \geq L_j. \quad (5.11)$$

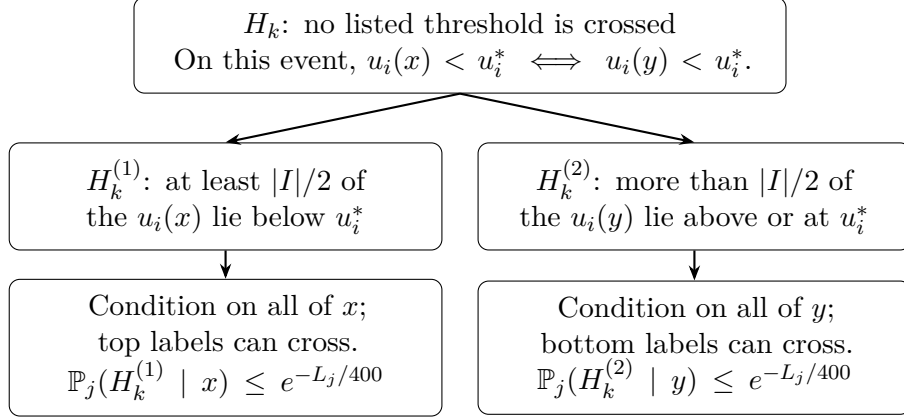


Figure 1: The two parts of the no-crossing event are small after conditioning on complementary endpoints. Each estimate controls the matching masked square sum in Lemma 5.1; no simultaneous avoidance over all indices is required.

For $w = b_s$ the first quantity is $\kappa_s m_s^{1/40} \geq L_s \geq L_j$; for $w = a_s$ it is $\kappa_s \sqrt{G_s} \geq L_{s-1} \geq L_j$.

Let H_k be the event of no crossing at any index in I , and define

$$H_k^{(1)} = H_k \cap \{\#\{i \in I : u_i(x) < u_i^*\} \geq |I|/2\}, \quad H_k^{(2)} = H_k \setminus H_k^{(1)}.$$

Conditional on x , the first event forbids the top label at every index in the indicated x -measurable set. Conditional independence and Lemma 4.4 therefore give

$$\mathbb{P}_j(H_k^{(1)} \mid x) \leq (1 - \rho_s(w)/10)^{|I|/2} \leq e^{-m_s \rho_s(w)/400} \leq e^{-L_j/400}.$$

If the x -measurable set has fewer than $|I|/2$ elements, the event is empty, so the same bound holds.

For the other conditional estimate, note explicitly that coordinate order and the absence of crossings imply

$$u_i(x) < u_i^* \iff u_i(y) < u_i^* \quad (i \in I).$$

Thus $H_k^{(2)}$ requires at least $|I|/2$ indices with $u_i(y) \geq u_i^*$. Conditional on y , the bottom labels at all those indices are forbidden. The same product estimate proves the second bound in (5.9). These arguments use the original law, without conditioning on E_j . Figure 1 displays the complementary split.

Outside H_k , at least one chosen coordinate crosses. All coordinates are ordered, so (3.7) and (5.11) give $R_{x,y}(k) \leq e^{-L_j}$. On H_k the ratio is at most 1. This proves (5.10) and the lemma. $\square$

5.6 Completion of the contradiction

Apply Lemma 5.1 to the labelled law $\mathbb{P}_j$, with

$$K = K_j, \quad D = D_j, \quad a = 1/j, \quad c = c_* = e^{-90}/2, \quad \eta = e^{-L_j}, \quad \delta = e^{-L_j/400}.$$

Equations (5.5) and (5.6) verify its algebraic assumptions. The proof of Lemma 5.3 supplies the event of mass at least c_* and all central-coefficient bounds for sufficiently large j . Lemma 5.4 gives the tail estimate with conditional probabilities under the full law. Consequently

$$c_* \varepsilon_j \leq (2K_j + 1)(e^{-L_j} + 2e^{-L_j/800}) \leq \frac{\varepsilon_j}{j},$$

where the last inequality is the parameter choice (2.10). Since $\varepsilon_j > 0$ and eventually $1/j < c_*$, this is impossible. This proves Theorem 5.2.

6 A continuous weight with one zero

The measurable obstruction of Theorem 5.2 is now in place. We will replace its logarithmic cost g_* by a function continuous off one point and tending to infinity at that point. The replacement changes the cost by the constant 1 plus a measurable coboundary. Since its transfer function need not be bounded, we begin with the precise operator statement that will preserve the obstruction.

6.1 Transport by an affiliated positive operator

A positive self-adjoint operator is *affiliated* with a von Neumann algebra if all its spectral projections belong to that algebra. Such an operator commutes, including on its domain, with every unitary in the commutant. The action of an invertible affiliated operator on projection lattices also appears in Mori [10, Proposition 3.1]. The form needed here includes a specified intertwining domain and preservation of the trace; we give a direct spectral-cutoff proof.

Lemma 6.1. *Let $(\mathcal{M}, \tau)$ be a finite von Neumann algebra acting on a Hilbert space, with faithful normal tracial state τ . Let H be an injective positive self-adjoint operator affiliated with $\mathcal{M}$, and let $A, B \in \mathcal{M}$. Suppose*

$$A(\text{Dom } H) \subseteq \text{Dom } H, \quad HA\zeta = BH\zeta \quad (\zeta \in \text{Dom } H).$$

For every projection $p \in \mathcal{M}$ with $A \text{ran } p \subseteq \text{ran } p$, the projection P onto

$$\overline{H(\text{ran } p \cap \text{Dom } H)}$$

belongs to $\mathcal{M}$, is invariant under B , and satisfies $\tau(P) = \tau(p)$.

Proof. Every commutant unitary preserves $\text{ran } p$ and $\text{Dom } H$, and commutes with H on that domain. It therefore maps the displayed closed space onto itself. Its projection commutes with the commutant, proving $P \in \mathcal{M}$. If $\zeta \in \text{ran } p \cap \text{Dom } H$, both invariance assumptions give $A\zeta \in \text{ran } p \cap \text{Dom } H$, and $BH\zeta = HA\zeta$. Thus B preserves the unclosed image and, by boundedness, its closure.

It remains to prove the trace assertion; bounded similarity cannot be assumed. Put $E_N = \mathbf{1}_{[1/N, N]}(H)$. Since H is injective, $E_N \uparrow 1$. For $q_N = p \wedge E_N$, the bounded element $Hq_N = (HE_N)q_N$ of $\mathcal{M}$ is bounded below by $1/N$ on its initial space. Its image is closed, and its polar decomposition identifies the initial projection q_N with its final projection, which is dominated by P . Therefore

$$\tau(P) \geq \tau(p \wedge E_N) \geq \tau(p) + \tau(E_N) - 1.$$

The last inequality follows directly from the polar decomposition of $(1 - E_N)p$: its initial support is $p - (p \wedge E_N)$ and its final support is at most $1 - E_N$. Letting N increase gives $\tau(P) \geq \tau(p)$.

For the reverse inequality, set $q'_N = (1 - p) \wedge E_N$. The bounded operator $H^{-1}q'_N = (H^{-1}E_N)q'_N$ is likewise bounded below on its initial space and has closed image. This image is orthogonal to $\text{ran } P$. Indeed, for $\eta \in \text{ran } q'_N$ and $\zeta \in \text{ran } p \cap \text{Dom } H$, self-adjointness gives

$$\langle H^{-1}\eta, H\zeta \rangle = \langle \eta, \zeta \rangle = 0;$$

here $H^{-1}\eta$ lies in $\text{Dom } H$. Polar decomposition and the same meet estimate yield

$$\tau(1 - P) \geq \tau((1 - p) \wedge E_N) \geq \tau(1 - p) + \tau(E_N) - 1.$$

Passing to the limit proves the opposite inequality for $\tau(P)$. □

6.2 Smoothing and moving the logarithmic drops

We claim that there exist a nonnegative function g , finite and continuous on $X \setminus \{0\}$ and tending to infinity at 0, and an almost everywhere finite real measurable function b such that

$$g = g_* + 1 + b \circ \sigma - b \quad \text{almost everywhere,} \quad \int_X g \, d\mu = \infty. \quad (6.1)$$

The construction has three parts: smooth each drop, move its support into a shrinking arc, and add an integrable background cost that guarantees divergence at the center of those arcs. Apart from the constant 1, all changes will be explicit coboundaries.

Fix $i \in J_j$ and put $\delta_i = \alpha_i/4$. For $u \in X$, define the continuous periodic smoothing of the sawtooth by

$$s_i(u) = \frac{1}{2\delta_i} \int_{-\delta_i}^{\delta_i} \{u + t\} \, dt.$$

It equals $\{u\}$ outside the δ_i -neighborhood of the cut at 0. Set

$$h_i(x) = C_j(\alpha_i + s_i(u_i(x)) - s_i(u_i(x) + \alpha_i)).$$

The elementary identity $\lfloor \{u\} + \alpha_i \rfloor = \alpha_i + \{u\} - \{u + \alpha_i\}$ shows that h_i is the corresponding average of the original one-coordinate cost. It is nonnegative and continuous, with

$$\int_X h_i \, d\mu = C_j \alpha_i, \quad \mu(\text{supp } h_i) \leq \alpha_i + 2\delta_i. \quad (6.2)$$

For

$$d_i^{\text{sm}}(x) = C_j(s_i(u_i(x)) - u_i(x)),$$

we have the exact measurable identity

$$h_i(x) = C_j \lfloor u_i(x) + \alpha_i \rfloor + d_i^{\text{sm}}(x) - d_i^{\text{sm}}(\sigma x). \quad (6.3)$$

The support of d_i^{sm} has measure at most $2\delta_i$.

Recall the shrinking arcs O_j , the integers h_j^* , and the continuous partition of unity $\phi_{j,n}$, $0 \leq n \leq h_j^*$, prepared in the parameter construction, with $\sigma^n(\text{supp } \phi_{j,n}) \subset O_j$. Define

$$h'_i(x) = \sum_{n=0}^{h_j^*} (h_i \phi_{j,n})(\sigma^{-n} x).$$

This is continuous and nonnegative, supported in O_j , and has the same integral as h_i . Telescoping gives

$$h'_i = h_i + d'_i - d'_i \circ \sigma, \quad d'_i(x) = \sum_{n=1}^{h_j^*} \sum_{r=1}^n (h_i \phi_{j,n})(\sigma^{-r} x). \quad (6.4)$$

Although this transfer contains a double sum, its support is contained in just h_j^* translates of $\text{supp } h_i$, one for each possible r . Consequently

$$\sum_i (\mu\{d_i^{\text{sm}} \neq 0\} + \mu\{d'_i \neq 0\}) \leq C \sum_j \frac{1 + h_j^*}{K_j} < \infty.$$

Thus each of the transfer sums has only finitely many nonzero terms almost everywhere, including at the translated point σx .

The sum $\sum_i h'_i$ is locally finite on $X \setminus \{0\}$ because its successive blocks are supported in $O_j \downarrow \{0\}$. It is therefore continuous there. Its integral is infinite: Tonelli's theorem and $\alpha_i > 1/(m_j K_j)$ give

$$\int_X \sum_i h'_i d\mu = \sum_j C_j \sum_{i \in J_j} \alpha_i \geq \sum_j C_j / K_j = \infty, \quad (6.5)$$

since $C_j \geq K_j^2$. Local finiteness alone need not force divergence at 0; the following background supplies that remaining property.

6.3 A background cost diverging at the zero

Let $W_j = 2^{j+1}$. Choose continuous functions $0 \leq v_j \leq 1$ with $v_j(0) = 1$, supported in closed arcs I_j shrinking to $\{0\}$. Choose these arcs small enough that $\sigma I_j, \dots, \sigma^{W_j} I_j$ are pairwise disjoint and

$$\sum_j W_j \mu(I_j) < \infty.$$

This is possible because each finite set of distinct points $\theta, 2\theta, \dots, W_j \theta$ has positive minimum separation. For $x \neq 0$, put

$$g_0(x) = 1 + \sum_j \left(v_j(x) - \frac{1}{W_j} \sum_{n=1}^{W_j} v_j(\sigma^{-n} x) \right).$$

For each j , disjointness bounds the norm of the subtracted function by $1/W_j$. These bounds sum to $1/2$, so the negative series converges uniformly. The positive series is locally finite away from 0. Consequently g_0 is continuous there and $g_0 \geq 1/2$. For every fixed integer J , the finite sum $\sum_{j \leq J} v_j(x)$ tends to J as $x \rightarrow 0$. The bounded negative series therefore implies $g_0(x) \rightarrow \infty$ as $x \rightarrow 0$. Its integral is finite, since $\sum_j \int v_j d\mu \leq \sum_j \mu(I_j) < \infty$ and the negative series is uniformly bounded.

Define

$$b_0(x) = \sum_j \frac{1}{W_j} \sum_{n=1}^{W_j} \sum_{r=1}^n v_j(\sigma^{-r} x).$$

The j th summand is supported in $\bigcup_{r=1}^{W_j} \sigma^r I_j$. The assumed summability makes this sum finite almost everywhere. On a conull set where the sums at x and σx are finite, telescoping shows

$$g_0 = 1 + b_0 \circ \sigma - b_0.$$

Now let

$$g(x) = g_0(x) + \sum_i h'_i(x) \quad (x \neq 0), \quad g(0) = \infty, \quad b = b_0 - \sum_i (d_i^{\text{sm}} + d'_i).$$

The preceding arguments prove that g is nonnegative, finite and continuous off 0, tends to infinity at 0, and has infinite integral. The transfer b is real and finite almost everywhere. Summing (6.3) and (6.4) is legitimate on the common conull set where the transfer sums and the original cost have finitely many nonzero terms. Together with the background identity this proves (6.1).

6.4 Conclusion and spectral consequences

Define the function on the unit circle by

$$f(e^{2\pi i x}) = e^{-g(x)}, \quad e^{-\infty} = 0.$$

It is continuous, is positive away from 1, and vanishes at 1. Its modulus is nonconstant and its zero set has Haar measure zero. Moreover,

$$\int_{\mathbb{T}} \log f \, dm = -\infty. \quad (6.6)$$

In particular, its Fuglede–Kadison determinant is zero [8]: the spectral distribution of $\pi(f)$ is the Haar distribution of f , so the determinant definition is precisely the exponential of (6.6). Also $|U\pi(f)| = \pi(f)$, which gives the same determinant for $T_f = U\pi(f)$. This conclusion uses the nonnegative integral calculation (6.5); it does not require the transfer b to be integrable.

Choose a finite Borel version of b and discard all integer translates of the null set where the coboundary identity fails. Let H_b be multiplication on the k th fiber coordinate by $e^{b(\sigma^k x)}$, with its natural square-integrability domain. It is positive, self-adjoint, and injective. Its spectral projections are π -images of spectral indicators of e^b , hence lie in M . Thus H_b is affiliated with M . The coboundary identity gives the pointwise factor equality

$$e^{b(\sigma^{k+1}x)} e^{-g(\sigma^k x)} = e^{-1} f_*(\sigma^k x) e^{b(\sigma^k x)}.$$

For every $\zeta \in \text{Dom } H_b$, the right-hand side defines the $k+1$ coordinate of $e^{-1} T_* H_b \zeta$, a square-integrable vector because T_* is bounded. Therefore

$$T_f \zeta \in \text{Dom } H_b, \quad H_b T_f \zeta = e^{-1} T_* H_b \zeta.$$

Lemma 6.1 sends any invariant projection of T_f to an invariant projection of $e^{-1} T_*$ having the same trace. Multiplication by the nonzero scalar e^{-1} does not change invariant ranges. Theorem 5.2 therefore excludes traces strictly between 0 and 1. This proves Theorem 1.1 for the arbitrary irrational θ fixed at the start.

The same continuous weight has a stronger spectral property, which we record with an elementary proof. This is the zero-determinant case of Zhu et al. [14, Theorem 2.3]. We include the truncated-logarithm argument, which uses uniform averaging under irrational rotation.

Corollary 6.2. *The constructed operator T_f is nonzero and quasinilpotent.*

Proof. For $M > 0$, the function $\ell_M(x) = \log \max(f(e^{2\pi i x}), e^{-M})$ is continuous. Its integrals decrease to $-\infty$ by (6.6). Irrational rotation averages of a continuous function converge uniformly to its integral: this follows for Fourier monomials from their geometric sums and then for continuous functions by uniform trigonometric approximation. Hence, for any $a > 0$, first choosing M with $\int_X \ell_M \, d\mu < -2a$ and then taking n sufficiently large gives

$$\sup_{x \in X} \prod_{k=0}^{n-1} f(e^{2\pi i \sigma^k x}) \leq \sup_{x \in X} \exp \left(\sum_{k=0}^{n-1} \ell_M(\sigma^k x) \right) \leq e^{-an}.$$

The expression on the left is $\|T_f^n\|$, because the power is a unitary U^n times multiplication by that continuous product. The spectral-radius formula yields radius zero, since a was arbitrary. Finally $T_f \neq 0$ because $f > 0$ almost everywhere. $\square$

A A product-rotation realization

The circle proof replaces independent coordinates by divisible characters and ordered digits. On a countable product of circles, independence is available directly. This gives a different sampling result: prescribed triangular laws in any finite set of coordinates are weak limits of positive orbit-pair measures, with both marginal densities uniformly bounded in L^2 . We prove that result and use it to construct a second concrete model for the convolution obstruction.

All notation in this appendix is local. Let

$$X = \mathbb{T}^{\mathbb{N}}, \quad \mathbb{T} = \mathbb{R}/\mathbb{Z},$$

with product Haar probability μ and its compact metrizable product topology. Coordinates are represented in $[0, 1)$, and $x \leq y$ means $x_i \leq y_i$ for every i . This order is Borel; its graph need not be closed at the circle cuts. For a sequence of angles $\theta = (\theta_i)$, write $\sigma_\theta x = x + \theta$. On $\mathcal{H} = L^2(X, \mu; \ell^2(\mathbb{Z}))$, put

$$(\pi(a)\zeta)(x)_k = a(\sigma_\theta^k x)\zeta(x)_k, \quad (U\zeta)(x) = S\zeta(x), \quad Se_k = e_{k+1},$$

and let $M = W^*(U, \pi(L^\infty(X)))$.

Theorem A.1 (Product realization). *There are angles $0 < \theta_i < 1$ and costs $c_i > 0$, with $\sum_i \theta_i < \infty$ and $1, \theta_1, \theta_2, \dots$ rationally independent, such that the measurable weight*

$$f(x) = \exp \left(- \sum_{i \geq 1} c_i [x_i + \theta_i] \right)$$

is positive almost everywhere and has $\int_X \log f \, d\mu = -\infty$. The algebra M is a type II_1 factor with separable predual, and $T = U\pi(f)$ is nonzero and norm-quasinilpotent, with $\Delta(T) = 0$. Every projection $p \in M$ satisfying $(1-p)Tp = 0$ is 0 or 1.

Moreover, $|T|$ has pure-point spectral type. The abelian algebra generated by any finite family of polar conjugates $U^n|T|U^{-n}$ is atomic, whereas $W^(U^n|T|U^{-n} : n \in \mathbb{Z})$ is diffuse.*

The last statement concerns spectral type: it does not require every point of the spectrum of $|T|$ to be an eigenvalue. In particular, zero can be a spectral accumulation point although $f > 0$ almost everywhere. As in the principal construction, the absence of invariant projections refers to the specified containing factor. For instance, the subspaces $L^2(X, \mu; \overline{\text{span}}\{e_k : k \geq m\})$ are invariant under the represented forward weighted shift, but their constant-field projections do not satisfy the covariance required for membership in M .

The proof has three product-specific tasks. We first choose angles for which the finite-coordinate sampling theorem holds. We then pass the invariant projection identity from those measures to biased product couplings. Finally we verify the positive-event and tail-attenuation hypotheses of Lemma 5.1. The field decomposition, compact-set transfer, and scalar sampling and density estimates are the lemmas already proved in the main text; no part of this appendix is needed for the prescribed-angle theorem.

A.1 Scales and independent angles

We choose the finite polynomial accuracy before choosing each new block. This leaves enough freedom to obtain both the sampling estimates and the threshold crossings needed by Lemma 5.1. All choices in this subsection are independent of any invariant projection.

Partition $\mathbb{N} = \{1, 2, \dots\}$ into consecutive finite blocks J_j of length m_j , writing

$$d_0 = 0, \quad d_j = \sum_{s \leq j} m_s, \quad J_j = \{d_{j-1} + 1, \dots, d_j\}.$$

The cutoff K_j will specify the finite coefficient window in the contradiction, while D_j and ε_j will give the coefficient bound and the product lower bound from Lemma 2.2. After these are fixed, L_j specifies the required decay of the omitted terms; the logarithmic drop C_j and the block length m_j are chosen to achieve that decay while retaining uniform bounds on the coupling marginals.

We now choose these parameters. Put $\kappa_j = 1/(100j)$ and start with $K_1 = 100$ and $G_1 = 1$. At stage j , the numbers K_j, G_j and all parameters of earlier blocks have already been fixed. Make the following choices in the stated order.

1. Define

$$D_j = \exp \left((K_j + 1) \sum_{s < j} C_s m_s \right).$$

Apply Lemma 2.2 with $K = K_j$, $D = D_j$, and $\delta = 1/j$ to choose $\varepsilon_j > 0$. This is possible because $D_j \geq 1 \geq 1/j$. Consequently,

$$\max_{0 \leq n \leq 2K_j} |(AB)_{-n}| \geq \varepsilon_j \quad (\text{A.1})$$

whenever A, B are supported in $[-K_j, K_j]$, their coefficients have modulus at most D_j , and $|A_0|, |B_0| \geq 1/j$.

2. Choose $L_j \geq 1$, increasing with j , so large that

$$(2K_j + 1)3 \exp(-L_j/10000) \leq \frac{\varepsilon_j}{j}. \quad (\text{A.2})$$

Set $C_j = \max\{L_j, K_j^2\}$, and choose a finite $G_{j+1} \geq 1$ such that

$$\kappa_{j+1} \sqrt{G_{j+1}} \geq L_j. \quad (\text{A.3})$$

3. Choose an integer $m = m_j$ sufficiently large, and set

$$K_{j+1} = \lceil m^{(3/4)m} \rceil,$$

subject to

$$\begin{aligned} m &\geq \max\{K_j, 2^j, 4G_j, 100\}, & d_{j-1} &\leq m/100, \\ \kappa_j m^{1/40} &\geq L_j, & G_{j+1} K_{j+1} &< m^{(4/5)m}. \end{aligned} \quad (\text{A.4})$$

Every choice is finite and feasible. Indeed, the first two steps do not depend on the new m_j . All but the final requirement in (A.4) are lower bounds on m . For the final one, the already fixed number G_{j+1} satisfies

$$\frac{G_{j+1} \lceil m^{(3/4)m} \rceil}{m^{(4/5)m}} \leq 2G_{j+1} m^{-m/20} \longrightarrow 0.$$

There is therefore no upper restriction conflicting with the lower bounds.

Write

$$N_j = m_j K_j, \quad a_j = G_j/m_j, \quad b_j = m_j^{-19/20}. \quad (\text{A.5})$$

The construction gives the useful consequences

$$\begin{aligned} m_j &\geq 2^j, & d_j &\leq 1.01m_j, & N_j &\leq m_j^2, \\ K_{j+1} &> 2K_j, & \sum_{s>j} \frac{1}{K_s} &\leq \frac{2}{K_{j+1}}, & \frac{1}{m_j} &\leq a_j, b_j \leq \frac{1}{4}. \end{aligned} \quad (\text{A.6})$$

For example, $m_j \geq 100$ and $m_j \geq K_j$ imply $K_{j+1} \geq m_j^{(3/4)m_j} > 2m_j \geq 2K_j$, and the reciprocal estimate follows by summing a geometric majorant. Also $m_{j+1} \geq K_{j+1} > m_j$, so each of m_j, K_j, N_j is increasing. The upper bound for a_j follows from $m_j \geq 4G_j$; the other width bounds follow from $G_j \geq 1$ and $m_j \geq 100$.

Use the coordinate order on X defined above.

On a separate product probability space choose independent random angles θ_i , with θ_i uniform on $[1/N_j, 2/N_j]$ when $i \in J_j$, and let

$$\sigma_\theta x = x + \theta \quad \text{in } X.$$

For every realization,

$$\sum_{i=1}^{\infty} \theta_i \leq 2 \sum_{j=1}^{\infty} \frac{m_j}{N_j} = 2 \sum_{j=1}^{\infty} \frac{1}{K_j} < \infty. \quad (\text{A.7})$$

Almost surely $1, \theta_1, \theta_2, \dots$ are rationally independent. To see this, multiply any proposed finite rational relation by a common denominator. If some angle has a nonzero integer coefficient, conditioning on all other angles leaves one prescribed value of an atomless random variable, an event of probability zero. A relation with no angle term cannot hold unless its constant term is zero. There are only countably many nontrivial finite integer relations.

For a rationally independent realization the rotation is ergodic and free. Indeed, the characters

$$x \mapsto \exp \left(2\pi i \sum_i v_i x_i \right), \quad v_i \in \mathbb{Z}, \quad v_i = 0 \text{ for all but finitely many } i,$$

form an orthonormal basis of $L^2(\mu)$: finite-coordinate functions are dense, and the usual Fourier bases are complete on each finite torus. Translation multiplies this character by $\exp(2\pi i \sum_i v_i \theta_i)$, which equals 1 only for the constant character. Hence the invariant L^2 functions are constant, which is ergodicity. Also $n\theta_1 \notin \mathbb{Z}$ for every $n \in \mathbb{Z} \setminus \{0\}$, so $\sigma_\theta^n x \neq x$ for every $x \in X$.

Proposition A.2 (Phase ranges). *Almost surely, for every sufficiently large j the following hold simultaneously for all integers k in the indicated ranges, with $m = m_j$:*

1. *If $G_j K_j \leq |k| \leq N_j/4$, then every $i \in J_j$ satisfies*

$$1 - \{k\theta_i\} \in [a_j, 1 - a_j].$$

2. *If $N_j/4 \leq |k| \leq m^{(4/5)m}$, then at least $m/20$ coordinates $i \in J_j$ satisfy*

$$1 - \{k\theta_i\} \in [b_j, 1 - b_j].$$

The first assertion in fact holds at every stage and for every realization of the angles in their prescribed intervals.

Proof. In the first range, put $t = |k|\theta_i$. The angle bounds give

$$a_j = G_j/m_j \leq t \leq 1/2.$$

For $k > 0$ the threshold is $1 - t$, and for $k < 0$ it is t . Both belong to $[a_j, 1 - a_j]$ because $a_j \leq 1/4$.

For the second range, the uniform law of $k\theta_i$ on an interval of length $|k|/N_j$ has, after reduction modulo one, a density bounded by

$$1 + N_j/|k| \leq 5.$$

This follows by counting the integer translates covering a point: the covering multiplicity of an interval of length r is at most $r + 1$. The statement also applies to negative k by reflection. Thus a given coordinate misses $[b_j, 1 - b_j]$ with probability at most $10b_j$. If fewer than $m/20$ coordinates satisfy the assertion, more than $19m/20$ coordinates miss the interval. Independence over $i \in J_j$ and a union bound over the possible sets of misses show that, for a fixed k , this failure probability is at most $2^m(10b_j)^{(19/20)m}$ for large m . A further union bound over the at most $2m^{(4/5)m}$ nonzero integers in the range gives

$$\mathbb{P}(\text{failure at stage } j) \leq 2m^{(4/5)m} 2^m (10b_j)^{(19/20)m} = \exp\left(-\frac{41}{400}m \log m + O(m)\right).$$

These probabilities are summable because $m_j \geq 2^j$. The first Borel–Cantelli Lemma now gives the second assertion for all sufficiently large j , simultaneously for every integer in each stated range. $\square$

We retain the random angles until the sampling construction below has supplied one further probability-one property.

A.2 Sampling ordered orbit pairs

We next approximate prescribed changes in finitely many coordinates by positive orbit moves. Both endpoint distributions must remain controlled: this will allow the projection identity to pass from orbit pairs to the couplings used in the product construction. The scalar estimates of Lemma 2.1 apply coordinate by coordinate; we prove here the growing-product estimate and the ordered weak limit.

The coordinate order is only Borel in the product-circle topology: already in one circle, the ordered pairs $(1/2, 1 - 1/n)$ converge to the unordered pair $(1/2, 0)$. Thus the closed-relation hypothesis in the classical coupling theorem [7, Corollary 3.2] does not apply directly. The result below also requires approximating measures supported on positive orbit pairs, with both marginal densities controlled; those are additional conclusions proved by the sampling construction.

For a finite set $I \subset \mathbb{N}$, define a finite Borel measure λ_I on $X \times X$ as follows. In each coordinate $i \in I$ use the unnormalized triangular measure

$$\mathbf{1}_{\{0 \leq x_i < y_i < 1\}} dx_i dy_i,$$

and in each other coordinate use the law of (x_i, x_i) for uniform $x_i \in [0, 1)$. Take the product over all coordinates. Equivalently, one can first normalize the finitely many triangular factors and then multiply the resulting probability measure by $2^{-|I|}$. In particular,

$$\lambda_I(X \times X) = 2^{-|I|}.$$

Its two marginal densities relative to μ are, respectively, $\prod_{i \in I} (1 - x_i)$ and $\prod_{i \in I} y_i$; they are bounded by 1. The measure is carried by $x \leq y$.

Proposition A.3 (Ordered orbit-pair sampling). *With probability one in the choice of θ , the following holds for every finite $I \subset \mathbb{N}$. There are positive finite Borel measures $\eta_{I,l}$ on $X \times X$ converging weakly to λ_I , each carried by ordered pairs of the form*

$$(x, y) = (x, \sigma_\theta^n x)$$

with n in a finite set of positive integers. Both marginals of $\eta_{I,l}$ have $L^2(\mu)$ densities, and their L^2 norms are bounded uniformly in l . The bound may depend on I and on the fixed realization of θ . Weak convergence here means convergence against every continuous function on the compact space $X \times X$.

Proof. We give the construction and its estimates before choosing the probability-one event. The notation $O(\cdot)$ in this proof has absolute constants, uniform in the coordinate, the selected powers, and the interval endpoints involved in a rectangle test.

We average many separated orbit powers so that both marginal densities approach those of an ideal model with independent uniform phases. The coordinate weights impose triangular laws on I and concentrate near the diagonal elsewhere, while their marginal bounds remain uniform in the growing product. A final restriction to ordered tail coordinates removes asymptotically negligible mass.

Separated powers and phase discrepancy Take l sufficiently large that $m_l \geq 200$, as required by the scalar lemma. At stage l write $m = m_l$, $d = d_l$, and set

$$Q = m^{(7/10)m}, \quad H = m^{300}, \quad R = N_l m^{1000}.$$

For all sufficiently large l we can choose, deterministically, a set $\mathcal{N}_l$ of integers in $[Q/2, Q]$ such that

$$P_l := |\mathcal{N}_l| \geq m^{0.69m}, \quad |vn + v'n'| > R \quad \left(\begin{array}{l} n, n' \in \mathcal{N}_l, \ n \neq n', \\ v, v' \in \mathbb{Z}, \ |v|, |v'| \leq H, \ (v, v') \neq (0, 0) \end{array} \right). \quad (\text{A.8})$$

Indeed, given a previously selected n , a fixed frequency pair with $v' \neq 0$ forbids an interval of possible n' of length $2R/|v'| \leq 2R$, hence at most $2R+2$ integers. There are $O(H^2)$ frequency pairs. Since $N_l \leq m^2$, each selection therefore excludes at most $O(m^{1602})$ integers. When $v' = 0$ and $v \neq 0$, the inequality is automatic because $n \geq Q/2 > R$ eventually; likewise, a zero first frequency creates no obstruction. The constraints are symmetric in the selected powers, since the full set of frequency pairs is used. Greedy selection from the $\asymp Q$ available integers yields at least $cQ/m^{1602} \geq m^{0.69m}$ choices for large m .

For a coordinate $i \in J_s$ with $s \leq l$, the angle θ_i is uniform on $[1/N_s, 2/N_s]$. Apply the separated-phase part of Lemma 2.1 with

$$N_0 = N_s \leq N_l = N \leq m^2.$$

The powers just selected satisfy its separation hypothesis. Thus $t_n = \{n\theta_i\}$ has a density $\varphi_{i,n}$ satisfying

$$\|\varphi_{i,n} - 1\|_\infty \leq N_s/n \leq N_l/n. \quad (\text{A.9})$$

In particular, that density is at most two for large l . For distinct selected powers and functions g, g' of bounded variation, the same lemma gives

$$\left| \mathbb{E}_\theta g(t_n) g'(t_{n'}) - \left(\int_0^1 g \right) \left(\int_0^1 g' \right) \right| \leq cm^{-80} (\|g\|_\infty + \text{Var } g) (\|g'\|_\infty + \text{Var } g'). \quad (\text{A.10})$$

No independence between t_n and $t_{n'}$ is asserted. The imported estimate uses their Fourier coefficients and one-dimensional marginal densities, and so applies even when their joint law is singular. We shall also use the direct consequence of (A.9)

$$\left| \mathbb{E}_\theta g(t_n) - \int_0^1 g(t) dt \right| \leq \frac{N_l}{n} \int_0^1 |g(t)| dt. \quad (\text{A.11})$$

The weighted measures and both marginal orientations Fix a finite I and restrict to sufficiently large l that $I \subset \{1, \dots, d\}$. Set

$$h = m^{-3/5}, \quad \Delta(s) = \min(s, 1-s), \quad \alpha = m^{-2}.$$

For $0 \leq s, t < 1$ define

$$w_i(s, t) = \begin{cases} \mathbf{1}_{\{0 < t < 1-s\}}, & i \in I, \\ \mathbf{1}_{\{\Delta(s) \geq \alpha\}} \frac{\mathbf{1}_{\{0 < t < h\Delta(s)\}}}{h\Delta(s)}, & i \notin I. \end{cases} \quad (\text{A.12})$$

The second line is defined to be zero if $\Delta(s) < \alpha$. Let $\tilde{\eta}_{I,l}$ be the average, over $n \in \mathcal{N}_l$, of the push-forward under $x \mapsto (x, \sigma_\theta^n x)$ of

$$W_n(x) d\mu(x), \quad W_n(x) = \prod_{i \leq d} w_i(x_i, \{n\theta_i\}). \quad (\text{A.13})$$

Each weight is bounded for fixed l , so these are finite positive measures. They are carried by pairs satisfying $x_i < y_i$ for $i \leq d$, since $h\Delta(x_i) \leq h(1 - x_i) < 1 - x_i$ and $h < 1$.

We will treat the marginals both without a test and after restriction to a cylinder rectangle

$$\mathcal{R} = \{(x, y) : x_i \in E_i^x, y_i \in E_i^y \text{ for every } i\},$$

where the E_i^x, E_i^y are intervals with rational endpoints and all but finitely many are the whole circle. Take l large enough that all tested coordinates are at most d . A missing test is represented by using the whole circle for every interval. For the first and second marginal, respectively, define

$$\begin{aligned} u_i^{(1)}(s, t) &= w_i(s, t) \mathbf{1}_{E_i^x}(s) \mathbf{1}_{E_i^y}(\{s + t\}), \\ u_i^{(2)}(s, t) &= w_i(\{s - t\}, t) \mathbf{1}_{E_i^x}(\{s - t\}) \mathbf{1}_{E_i^y}(s). \end{aligned} \quad (\text{A.14})$$

Changing variables $y = \sigma_\theta^n x$ preserves Haar measure. It follows that the two tested marginal densities are exactly the averages of the products of the corresponding $u_i^{(r)}$. There is no factor from coordinates past d , even in the second orientation, since their unweighted Haar law is translation invariant.

We check the extra factors introduced by triangular laws and interval tests. Write u_i for either orientation and put

$$\bar{u}_i(s) = \int_0^1 u_i(s, t) dt, \quad A_i = \int_0^1 \bar{u}_i(s)^2 ds.$$

Uniformly in s , the coordinate, and the test intervals, we have

$$\|u_i(s, \cdot)\|_\infty + \text{Var}_t u_i(s, \cdot) = O(m^{13/5}) = O(m^{10}). \quad (\text{A.15})$$

For an untested triangular factor, the bound is $O(1)$: in the first orientation its support in t is $0 < t < 1 - s$, and in the second it is $0 < t \leq s$. The possible value at $t = s$ affects no integral. For the untested near-diagonal factors, Lemma 2.1 gives the bound $O(m^{13/5})$. To include the tests, in either orientation the extra factor is a constant in $\{0, 1\}$ times an interval indicator translated or reflected modulo one. Its supremum is at most one and its variation is bounded by an absolute constant. The inequality

$$\text{Var}(ab) \leq \|a\|_\infty \text{Var}(b) + \|b\|_\infty \text{Var}(a)$$

therefore proves (A.15), including all jumps at the test endpoints.

The moment bounds needed below are

$$0 \leq \bar{u}_i(s) \leq 2, \quad 0 \leq A_i \leq 1 + O(h^2), \quad \int_0^1 \int_0^1 u_i(s, t)^2 ds dt = O(h^{-1} \log m). \quad (\text{A.16})$$

For an untested triangular factor the two averages are exactly $1 - s$ and s , respectively. In either orientation its total integral and squared integral are $1/2$, while $A_i = 1/3$. For an untested near-diagonal factor, Lemma 2.1 gives

$$\|\bar{u}_i\|_\infty \leq 2, \quad A_i \leq 1 + \frac{h^2}{(1-h)^2}, \quad \iint u_i(s, t)^2 ds dt = \frac{2}{h} \log \frac{1}{2\alpha}.$$

Since tests only decrease the nonnegative factors and their averages, these statements imply (A.16) for every tested factor. Their actual masses need not agree; in particular, the near-diagonal integral identities are not being imposed on the triangular factors or on the tested kernels. The quadratic error $O(h^2)$ is what keeps the product of the moment bounds controlled as the number of coordinates grows.

The estimate in the growing product dimension For either fixed orientation and a fixed rectangle test or no test, write

$$\begin{aligned} q_n(z) &= \prod_{i \leq d} u_i(z_i, \{n\theta_i\}), & Z_l(z) &= \frac{1}{P_l} \sum_{n \in \mathcal{N}_l} q_n(z), \\ Z_l^0(z) &= \prod_{i \leq d} \bar{u}_i(z_i), & A &= \prod_{i \leq d} A_i = \|Z_l^0\|_2^2. \end{aligned}$$

We prove the estimate

$$\mathbb{E}_\theta \|Z_l - Z_l^0\|_{L^2(\mu)}^2 \leq O(dm^{-40}) + \frac{(ch^{-1} \log m)^d}{P_l}. \quad (\text{A.17})$$

The independence used here is independence across coordinates θ_i . In a single coordinate two phases may still be dependent, as already accounted for in (A.10).

For distinct n, n' , Tonelli's Theorem, product Haar measure, and coordinate independence yield the exact factorization

$$\mathbb{E}_\theta \langle q_n, q_{n'} \rangle = \prod_{i \leq d} \int_0^1 \mathbb{E}_\theta [u_i(s, \{n\theta_i\}) u_i(s, \{n'\theta_i\})] ds.$$

By (A.15) and (A.10), its i th factor is $A_i + e_i$ with $|e_i| \leq cm^{-50}$. Similarly,

$$\mathbb{E}_\theta \langle q_n, Z_l^0 \rangle = \prod_{i \leq d} \int_0^1 \bar{u}_i(s) \mathbb{E}_\theta u_i(s, \{n\theta_i\}) ds.$$

Here (A.11) and $\bar{u}_i \leq 2$ bound the error in each factor from A_i by $4N_l/n = O(m^{-50})$. Thus both sorts of factors have the same stated error bound.

The product errors do not require any lower bound on A_i . Telescoping gives

$$\left| \prod_{i \leq d} (A_i + e_i) - \prod_{i \leq d} A_i \right| \leq cdm^{-50} (1 + ch^2 + cm^{-50})^{d-1} = O(dm^{-50}), \quad (\text{A.18})$$

because (A.6) gives $d \leq 1.01m$, and hence $dh^2 \leq 1.01m^{-1/5}$. In particular, $A \leq (1 + ch^2)^d = O(1)$.

For a self-pairing, use only the one-phase density bound at most 2 and (A.16):

$$\mathbb{E}_\theta \|q_n\|_2^2 = \prod_{i \leq d} \int_0^1 \mathbb{E}_\theta u_i(s, \{n\theta_i\})^2 ds \leq (ch^{-1} \log m)^d.$$

To see the cancellation of all other main terms explicitly, expand the squared norm. The distinct-pair terms have total averaging weight $1 - 1/P_l$ and main term $(1 - 1/P_l)A$. The cross terms with Z_l^0 have main term $-2A$, and the final squared norm of Z_l^0 is A . Their sum is $-A/P_l \leq 0$. By (A.18), their remaining contribution has absolute value $O(dm^{-50})$. The self-pairings have total weight $1/P_l$. Dropping $-A/P_l$ proves the slightly weaker bound (A.17).

This estimate is summable in l . Indeed,

$$\log_m(ch^{-1} \log m)^d \leq 1.01m \left(\frac{3}{5} + \log_m(c \log m) \right) \leq 0.65m$$

for large m , whereas $P_l \geq m^{0.69m}$. Thus

$$\mathbb{E}_\theta \|Z_l - Z_l^0\|_2^2 \leq O(m^{-39}) + m^{-0.04m}, \quad (\text{A.19})$$

whose sum is finite by $m_l \geq 2^l$.

One probability-one event and the weak limit For each fixed finite I , orientation, and rational cylinder rectangle or absent test, Tonelli's Theorem applied to (A.19) shows that

$$\sum_l \|Z_l - Z_l^0\|_2^2 < \infty \quad \text{almost surely,}$$

where only the sufficiently large admissible stages are included. In particular, $\|Z_l - Z_l^0\|_2 \rightarrow 0$. Finite subsets of $\mathbb{N}$ form a countable family, as do finite cylinder rectangles with rational endpoints, and there are only two orientations. Intersect these countably many probability-one events. For the remainder of the proof take any θ in this intersection.

Without a test, Z_l is a marginal density of $\tilde{\eta}_{I,l}$. Since $\|Z_l^0\|_2^2 = A = O(1)$, the L^2 convergence just proved bounds both marginal norms uniformly in l . We may omit finitely many initial stages; alternatively, each of their densities is bounded for fixed l , so they can be included by enlarging the bound. The total masses of $\tilde{\eta}_{I,l}$ are consequently uniformly bounded.

For a fixed cylinder rectangle $\mathcal{R}$, using the first orientation with its test shows that

$$\left| \tilde{\eta}_{I,l}(\mathcal{R}) - \int_X Z_l^0 d\mu \right| \leq \|Z_l - Z_l^0\|_2 \rightarrow 0.$$

The ideal integral has a direct product interpretation on the first d coordinates. For $i \in I$, integration of $w_i(x_i, t)$ in x_i, t gives exactly the triangular measure under $y_i = x_i + t$. For $i \notin I$, first take uniform x_i , discard $\Delta(x_i) < \alpha$, and, otherwise, take t uniform in $(0, h\Delta(x_i))$, setting $y_i = x_i + t$. Before the discard this is a probability law (the value at the two endpoint values of x_i may be defined arbitrarily), and $|y_i - x_i| \leq h/2$. The total loss caused by all cutoffs is at most

$$2d\alpha = 2dm^{-2} \rightarrow 0,$$

by the union bound, with the finite triangular factors having mass at most one. Each fixed tested coordinate outside I therefore converges to its uniform diagonal law. More explicitly, outside an h -neighborhood of a tested interval's endpoints, membership of y_i in that interval agrees with membership of x_i ; the neighborhood has Lebesgue measure $O(h)$. Since a fixed rectangle tests only finitely many coordinates, all such errors vanish. Untested coordinates contribute only their masses, and the total cutoff estimate controls their product even as d increases. It follows that

$$\tilde{\eta}_{I,l}(\mathcal{R}) \rightarrow \lambda_I(\mathcal{R}). \quad (\text{A.20})$$

These limits imply weak convergence on $X \times X$. Here are the topological details. The boundaries of the interval rectangles have zero λ_I measure, since every single-coordinate marginal of λ_I is absolutely continuous. Every continuous function on the compact product can be uniformly approximated by a continuous function of finitely many coordinates: in a summable product metric, replace the tail coordinates by fixed points and use uniform continuity. On the resulting finite torus, a sufficiently fine rational grid gives a step function, a finite linear combination of rational rectangle indicators, with arbitrarily small uniform error. Thus (A.20) and the uniform bound on total masses give convergence of integrals for every continuous function. In particular,

$$\tilde{\eta}_{I,l} \implies \lambda_I.$$

Trimming the tail to impose order everywhere The untrimmed measures now have the required weak limit and both marginal bounds. It remains to impose order past the first d coordinates. Define $\eta_{I,l}$ by restricting $\tilde{\eta}_{I,l}$ to the Borel set where $x_i \leq y_i$ also for every $i > d$. For a fixed selected n , the weight W_n depends only on the first d coordinates, so the tail x -coordinates remain independent uniform coordinates under that summand, apart from multiplication by its total mass. For a uniform tail coordinate, the probability that $x_i > \{x_i + n\theta_i\}$ is exactly $\{n\theta_i\} \leq n\theta_i$. Consequently, uniformly over $n \in \mathcal{N}_l$, the relative mass discarded from this summand is at most

$$\begin{aligned} \sum_{i>d} n\theta_i &\leq 2m^{(7/10)m} \sum_{s>l} \frac{1}{K_s} \\ &\leq \frac{4m^{(7/10)m}}{K_{l+1}} \leq 4m^{-m/20} \longrightarrow 0. \end{aligned} \tag{A.21}$$

The union bound is valid for the countable tail by increasing finite unions. More precisely, the discarded mass of a summand equals its total mass times the Haar probability of its tail failure event, because the weight and that event depend on disjoint coordinates. This statement also holds for a summand of mass zero. Averaging over n shows that the total discarded mass is bounded by (A.21) times the uniformly bounded untrimmed mass. Hence the difference of the two measures tends to zero in total variation and $\eta_{I,l} \implies \lambda_I$.

Each trimmed marginal is a submeasure of the corresponding untrimmed marginal. Its Radon–Nikodym density is therefore nonnegative and bounded almost everywhere by the untrimmed density, so its L^2 norm cannot increase. Trimming preserves the finite set of positive orbit powers and now imposes order in every coordinate. All conclusions hold on the single probability-one event already constructed, for every finite I , proving the Proposition. $\square$

We now fix one realization θ for which rational independence, Proposition A.2, and Proposition A.3 all hold, and henceforth write $\sigma = \sigma_\theta$. The scales and this rotation are fixed before any candidate invariant projection is introduced. For this fixed rotation, each sampling sequence has actual weak convergence against every continuous function; its validity is no longer restricted to the countable tests used to obtain the probability-one event. Thus later choices of continuous tests or compact sets depending on a candidate projection require no further intersection of events in the angle probability space.

A.3 The product factor and its weighted shift

We now turn the fixed rotation into an operator in a finite factor. The goal is to express invariance as an identity on ordered pairs, where the sampling result can act. Use the rotation σ fixed at the end of Section A.2, and write $\mathcal{K} = \ell^2(\mathbb{Z})$. The operators $U, \pi(g)$ and their

generated algebra M are those defined at the beginning of the appendix. This is the usual regular representation of a crossed product by a measure-preserving transformation [1, §1.4 and Proposition 1.4.5]; the class of operators $U\pi(f)$ is also considered in [6, Section 5]. For completeness, we verify the properties of M needed here.

Lemma A.4. *The algebra M is a factor of type II_1 with separable predual. Its elements are essentially bounded measurable fields $H(x) \in \mathcal{B}(\mathcal{K})$ satisfying*

$$H(\sigma x) = S^{-1}H(x)S \quad \text{almost everywhere,} \quad (\text{A.22})$$

and its faithful normal tracial state is

$$\tau(H) = \int_X H(x)_{0,0} d\mu(x).$$

Proof. Each generator commutes with scalar multiplication by any function of x . It also commutes with the unitary $(J\zeta)(x) = S\zeta(\sigma x)$. These commutation relations therefore hold for every $H \in M$. Lemma 3.1 therefore represents H by an essentially bounded measurable field $H(x)$, with products and adjoints computed fiberwise. Commutation with J gives (A.22).

Conversely, let H be an essentially bounded measurable field satisfying (A.22). All its integer-iterate covariance identities hold on one conull set. Put $h_r(x) = H(x)_{r,0}$ for $r \in \mathbb{Z}$. Then

$$H(x)_{k+r,k} = h_r(\sigma^k x),$$

so the r th matrix diagonal of H is $U^r \pi(h_r) \in M$. To recover H from these diagonals, use the constant-field unitaries $D_t e_k = e^{2\pi i k t} e_k$, $t \in \mathbb{T}$. The strong integral

$$H_N = \int_0^1 \mathcal{F}_N(t) D_t H D_t^* dt = \sum_{|r| < N} \left(1 - \frac{|r|}{N}\right) U^r \pi(h_r)$$

belongs to M , where $\mathcal{F}_N$ is the Fejér kernel. The map $t \mapsto D_t H D_t^*$ is strongly continuous and uniformly bounded. The positivity, unit integral and concentration at zero of $\mathcal{F}_N$ imply $H_N \rightarrow H$ strongly. The strong closure of M therefore proves the converse field characterization.

The displayed state is the vector state associated with the constant field e_0 , hence is normal. If $H \geq 0$ and $\tau(H) = 0$, its 00 entry vanishes almost everywhere. Covariance then forces every diagonal entry to vanish almost everywhere. Positivity, or the inequality $|H_{k,l}|^2 \leq H_{k,k} H_{l,l}$, implies $H = 0$. Thus the state is faithful. For $H, H' \in M$, covariance and the change of variable $z = \sigma^r x$ give

$$\int_X H(x)_{0,r} H'(x)_{r,0} d\mu(x) = \int_X H'(z)_{0,-r} H(z)_{-r,0} d\mu(z).$$

The sums over r are absolutely integrable by the row and column ℓ^2 bounds and Cauchy–Schwarz. Summing proves $\tau(HH') = \tau(H'H)$.

The rotation σ is ergodic by Section A.1. If H is central, it commutes with $\pi(x \mapsto e^{2\pi i x_1})$. In each fiber this last operator has distinct diagonal entries, because θ_1 is irrational; hence $H(x)$ is diagonal. Commutation with U makes all its diagonal entries equal. Equation (A.22) and ergodicity make their common value constant almost everywhere. The center of M is therefore scalar. The algebra contains the infinite-dimensional abelian algebra $\pi(L^\infty(X))$, so this finite factor is not a matrix algebra and is of type II_1 . Finally, the predual of a von Neumann algebra on a separable Hilbert space is a quotient of the separable trace class, and is separable. $\square$

Weights and cocycles

Using representatives $x_i \in [0, 1)$ for the circle coordinates, put

$$f(x) = \exp \left(- \sum_{j \geq 1} C_j \sum_{i \in J_j} \lfloor x_i + \theta_i \rfloor \right), \quad T = U\pi(f). \quad (\text{A.23})$$

Since $\sum_i \theta_i \leq 2 \sum_j K_j^{-1} < \infty$, the first Borel–Cantelli Lemma shows that only finitely many terms in this sum are nonzero almost everywhere. Thus $0 < f \leq 1$ almost everywhere and T is a nonzero bounded operator. For every $k \in \mathbb{Z}$ define

$$F_k(x) = \exp \left(- \sum_{j \geq 1} C_j \sum_{i \in J_j} \lfloor x_i + k\theta_i \rfloor \right).$$

For a fixed k , all sufficiently late coordinates have $|k|\theta_i < 1$, and the probability that their floor is nonzero is $|k|\theta_i$. Borel–Cantelli again proves that F_k is positive and finite almost everywhere. Taking a countable intersection and all its σ -translates gives an invariant Borel set X_0 of full measure on which all these formulas hold and

$$F_0 = 1, \quad F_{k+n}(x) = F_n(\sigma^k x) F_k(x) \quad (k, n \in \mathbb{Z}). \quad (\text{A.24})$$

Indeed, the identity follows coordinatewise from $\lfloor x_i + (k+n)\theta_i \rfloor = \lfloor x_i + k\theta_i \rfloor + \lfloor \{x_i + k\theta_i\} + n\theta_i \rfloor$. Extend each F_k to be positive and finite on $X \setminus X_0$; its values there will never affect an integral.

For $x, y \in X_0$ with $x \leq y$ coordinatewise, set

$$R_{x,y}(k) = \frac{F_k(y)}{F_k(x)} \leq 1, \quad k \in \mathbb{Z}. \quad (\text{A.25})$$

The inequality holds for negative as well as positive k , since the floor function is nondecreasing. More precisely, each coordinate $i \in J_s$ with

$$x_i < 1 - \{k\theta_i\} \leq y_i \quad (\text{A.26})$$

contributes a factor e^{-C_s} to this ratio. All other coordinates contribute either 1 or another such factor.

Lemma A.5. *The weight satisfies $\int_X \log f \, d\mu = -\infty$ and $\Delta(T) = 0$. The operator T is nonzero and norm-quasinilpotent: $\lim_{n \rightarrow \infty} \|T^n\|^{1/n} = 0$.*

Proof. For $n \geq 1$, direct multiplication and (A.24) give $T^n = U^n \pi(F_n)$. All the floors defining F_n are nonnegative. For each fixed q , the estimate $\lfloor x_i + n\theta_i \rfloor \geq n\theta_i - 1$ on the first q blocks therefore gives

$$\limsup_{n \rightarrow \infty} \|T^n\|^{1/n} \leq \exp \left(- \sum_{s \leq q} C_s \sum_{i \in J_s} \theta_i \right) \leq \exp \left(- \sum_{s \leq q} \frac{C_s}{K_s} \right).$$

Since $C_s \geq K_s^2$, the last expression tends to zero as $q \rightarrow \infty$. Also, Tonelli's Theorem and the same angle bounds give

$$- \int_X \log f \, d\mu = \sum_s C_s \sum_{i \in J_s} \theta_i \geq \sum_s \frac{C_s}{K_s} = \infty.$$

Since $|T| = \pi(f)$ and the trace on $\pi(L^\infty(X))$ is integration against μ , the determinant formula gives $\Delta(T) = 0$. Positivity of f almost everywhere ensures $T \neq 0$. $\square$

The uniform norm estimate also gives $\sigma(T) = \{0\}$ and Brown measure δ_0 .

From orbit identities to triangular laws

The sampled measures now allow us to use Lemma 3.3. The order supplies its uniform cap, and the two marginal L^2 bounds supply uniform absolute continuity.

Proposition A.6. *Suppose $p \in M$ is a projection and $(1 - p)Tp = 0$. For every finite $I \subset \mathbb{N}$,*

$$(1 - p(y)) \operatorname{diag}_{k \in \mathbb{Z}}(R_{x,y}(k))p(x) = 0 \quad \text{for } \lambda_I\text{-almost every } (x, y). \quad (\text{A.27})$$

Proof. Fix this p and a Borel version of its projection field. We may shrink X_0 to a common invariant Borel set of full measure on which covariance, the cocycle identities, and all the following countably many power identities hold. Invariance of $\operatorname{ran} p(x)$ under $T(x)$ gives invariance under $T(x)^n$, for each $n \geq 1$. Applying S^{-n} and using covariance yields

$$(1 - p(\sigma^n x)) \operatorname{diag}_{k \in \mathbb{Z}}(F_n(\sigma^k x))p(x) = 0. \quad (\text{A.28})$$

Division by $F_n(x) > 0$ changes the diagonal entries to $F_k(\sigma^n x)/F_k(x)$ by (A.24). Proposition A.3 gives ordered orbit-pair measures converging to λ_I with uniformly L^2 -bounded marginal densities. Their first marginals are absolutely continuous, so the countable collection of orbit identities holds almost everywhere on each approximant. The marginals of λ_I are absolutely continuous. The uniform L^2 bounds give uniform absolute continuity by Cauchy–Schwarz. Orderedness and (A.25) give the common cap $B = 1$ on every ratio, under both the approximants and the limit. Lemma 3.3 now proves (A.27). $\square$

For later use, the common strict-diagonal lemma applies here: the base is ergodic, the integral trace is faithful, and $f > 0$ almost everywhere. Thus, for any projection $p \in M$ with $0 < p < 1$ and $(1 - p)Tp = 0$, Lemma 3.2 gives

$$0 < r(x) := p(x)_{0,0} < 1 \quad \text{for almost every } x. \quad (\text{A.29})$$

A.4 Biased couplings of product coordinates

The measures λ_I give the pair identity when only finitely many coordinates are replaced. We now construct ordered probability laws that give every block opportunities to cross its thresholds. Their endpoint densities remain uniformly bounded in $L^2(\mu)$, and the crossing estimates remain valid after either entire endpoint is fixed. These are the properties needed for the convolution obstruction.

An ordered coupling with both marginals equal to Haar measure would give $x = y$ almost surely: in every coordinate, $y_i - x_i \geq 0$ and $\mathbb{E}(y_i - x_i) = 0$. We therefore allow a controlled change in the endpoint marginals. Fix a block index $j \geq 1$. For $i \in J_s$, write $m = m_s$ and set

$$\rho_s(w) = \kappa_s \sqrt{w/m}, \quad w = a_s, b_s. \quad (\text{A.30})$$

Choose the coordinate pair (x_i, y_i) by the following mixture:

- For each of the two labelled widths $w = a_s, b_s$, include a top component of probability $\rho_s(w)$. In this component x_i and y_i are independent uniforms on $[0, 1 - w/2)$ and $[1 - w/2, 1)$, respectively.
- For each labelled width $w = a_s, b_s$, include a bottom component of probability $\rho_s(w)$, with independent uniforms on $[0, w/2)$ and $[w/2, 1)$, respectively.
- If $s = j$, include a tagged component of probability $1/m$, with independent uniforms on $[0, 1/m)$ and $[1 - 1/m, 1)$, respectively.

- With the remaining probability

$$q_{j,s} = 1 - 2\rho_s(a_s) - 2\rho_s(b_s) - \mathbf{1}_{\{s=j\}}/m, \quad (\text{A.31})$$

choose $x_i = y_i$ uniformly in $[0, 1)$.

The width labels remain distinct even if their numerical values agree. The scale bounds imply

$$\frac{1}{m} \leq a_s, b_s \leq \frac{1}{4}, \quad \frac{\rho_s(w)}{w} = \frac{\kappa_s}{\sqrt{mw}} \leq \kappa_s \leq \frac{1}{100}. \quad (\text{A.32})$$

Since $m \geq 100$, it follows that $q_{j,s} \geq 1 - \kappa_s - 1/m \geq 0.98$.

Make the choices independently over all coordinates. More precisely, let $(\Omega_j, \mathbb{P}_j)$ be the countable product of the spaces carrying the mixture label and its coordinate pair. Let $x, y : \Omega_j \rightarrow X$ be the endpoint maps, and let ν_j be the law of (x, y) on X^2 . The labels, including the tags, are thus retained as events on Ω_j . These are the actual product coordinates of the endpoints. Each component is supported on $x_i \leq y_i$, so $x \leq y$ almost surely.

Lemma A.7 (Marginals and conditional crossings). *Put*

$$H^2 = \exp \left(20 \left(2 \sum_{s \geq 1} \kappa_s^2 + 1 \right) \right) < \infty. \quad (\text{A.33})$$

For every $j \geq 1$, the law just constructed has the following properties.

1. Every one-coordinate endpoint density is at most 10. Both marginals of ν_j have $L^2(\mu)$ densities of norm at most H . The same bound holds for the product law formed by any deterministic choice of the x -marginal or the y -marginal at each coordinate.
2. Conditional on x , the coordinates y_i are independent. Conditional on y , the coordinates x_i are independent. If $i \in J_s$, $w = a_s$ or b_s , and $u \in [w, 1 - w]$, then

$$\mathbb{P}_j(x_i < u \leq y_i \mid x) \geq \rho_s(w)/10 \quad \text{on } \{x_i < u\}, \quad (\text{A.34})$$

$$\mathbb{P}_j(x_i < u \leq y_i \mid y) \geq \rho_s(w)/10 \quad \text{on } \{y_i \geq u\}. \quad (\text{A.35})$$

The assertions hold almost surely, simultaneously for any specified countable family of such thresholds.

Proof. For a top or bottom component of width w , each endpoint interval has length at least $w/2$, so its density is at most $2/w$. A tagged endpoint density is at most m . Consequently either one-coordinate marginal density h_i satisfies

$$0 < q_{j,s} \leq h_i \leq 1 + 4 \sum_{w=a_s, b_s} \frac{\rho_s(w)}{w} + \mathbf{1}_{\{s=j\}} \leq 2 + 8\kappa_s < 10.$$

Throughout these sums the two widths are counted with their labels.

If v is the uniform density on an interval of length at least $w/2$, then $\int(v-1) = 0$ and $\|v-1\|_2^2 \leq 2/w$. The corresponding bound for a tagged interval is m . Subtracting 1 from h_i expresses it as the sum of four weighted mean-zero perturbations and, when $s = j$, one tagged perturbation. Cauchy-Schwarz for these at most five summands gives

$$\begin{aligned} \|h_i - 1\|_2^2 &\leq 20 \sum_{w=a_s, b_s} \frac{\rho_s(w)^2}{w} + 5 \frac{\mathbf{1}_{\{s=j\}}}{m_s} \\ &\leq 20 \left(\frac{2\kappa_s^2}{m_s} + \frac{\mathbf{1}_{\{s=j\}}}{m_s} \right). \end{aligned} \quad (\text{A.36})$$

This estimate holds for either endpoint. Summing over the m_s coordinates of every block gives the bound $20(2\sum_s \kappa_s^2 + 1)$ for every deterministic selection of endpoint marginals. Each selected coordinate density is nonnegative and has integral one, and the coordinate laws are independent. Thus Lemma 4.1 supplies the asserted L^2 density with norm at most H . In particular, for each such product law η and every Borel set $A \subseteq X$,

$$\eta(A) \leq H\mu(A)^{1/2}. \quad (\text{A.37})$$

To verify the assertions after conditioning on an entire endpoint, we describe the conditional kernels. At coordinate $i \in J_s$, let ℓ run through the replacement labels, including the tag when present. Write π_ℓ for its mixture probability and u_ℓ, v_ℓ for the normalized interval densities of its two endpoints. The marginal densities in this coordinate are

$$h_i^x(t) = q_{j,s} + \sum_\ell \pi_\ell u_\ell(t), \quad h_i^y(t) = q_{j,s} + \sum_\ell \pi_\ell v_\ell(t).$$

They are strictly positive. A conditional law of y_i given $x_i = t$ is

$$K_i^x(t, dz) = \frac{q_{j,s}}{h_i^x(t)} \delta_t(dz) + \sum_\ell \frac{\pi_\ell u_\ell(t)}{h_i^x(t)} v_\ell(z) dz. \quad (\text{A.38})$$

The reverse kernel K_i^y interchanges u_ℓ and v_ℓ . Independence of the coordinate pairs shows, first for cylinder tests, that $\bigotimes_i K_i^x(x_i, \cdot)$ is a conditional law of y given x . The monotone class theorem extends the defining identity to all Borel tests on X^2 . The corresponding product of reverse kernels is a conditional law of x given y . These formulas prove both assertions of conditional independence.

If $x_i = t < u \leq 1 - w$, then $t \in [0, 1 - w/2)$, where the top component of width w has x_i -density $(1 - w/2)^{-1} \geq 1$. Its whole y_i interval lies above u . Since $h_i^x \leq 10$, its contribution to (A.38) is at least $\rho_s(w)/10$. This proves (A.34). If $y_i = t \geq u \geq w$, the bottom component has y_i -density $(1 - w/2)^{-1} \geq 1$ at t , and its whole x_i interval lies below u . The reverse kernel therefore proves (A.35). Intersecting the full-measure sets for any specified countable family of thresholds gives the final assertion. $\square$

It remains to establish the pair identity under these probability laws. We first replace a finite prefix of x by coordinates of y . The resulting law is dominated by a finite sum of the triangular laws λ_I , but its diagonal tail retains the biased x -distribution. Keeping that tail density explicit permits the transfer to infinitely many replacements.

Proposition A.8 (The identity under the biased couplings). *Let $p \in M$ be an orthogonal projection satisfying $(1 - p)Tp = 0$ in the product model. For every $j \geq 1$,*

$$(1 - p(y)) \text{diag}_{k \in \mathbb{Z}}(R_{x,y}(k))p(x) = 0 \quad \text{for } \nu_j\text{-almost every } (x, y). \quad (\text{A.39})$$

Moreover, $0 < R_{x,y}(k) \leq 1$ for all $k \in \mathbb{Z}$ almost surely.

Proof. Fix j . For $D \geq 1$, define

$$y_i^{(D)} = \begin{cases} y_i, & i \leq D, \\ x_i, & i > D, \end{cases}$$

and let β_D be the law of $(x, y^{(D)})$ under $\mathbb{P}_j$. We claim that

$$\beta_D \ll \sum_{I \subseteq \{1, \dots, D\}} \lambda_I. \quad (\text{A.40})$$

Condition on a pattern c of mixture labels in the first D coordinates. There are finitely many patterns. Let $I(c)$ consist of the coordinates carrying replacement labels. At each $i \in I(c)$ the conditioned pair has density

$$r_{c,i}(t, z) = \frac{\mathbf{1}_{U_{c,i}}(t) \mathbf{1}_{V_{c,i}}(z)}{|U_{c,i}| |V_{c,i}|}$$

relative to the unnormalized triangular measure $\mathbf{1}_{\{0 \leq t < z < 1\}} dt dz$. Here $U_{c,i}, V_{c,i}$ are the two intervals of the selected component. Their lengths are positive and their rectangle is supported on $t < z$. The remaining prefix coordinates are uniformly diagonal.

For the diagonal tail, let g be the full x -marginal density from Lemma A.7. Write $\mu = \mu_{\leq D} \otimes \mu_{>D}$ and define

$$g_D(z) = \int g(u, z) d\mu_{\leq D}(u), \quad \|g_D\|_{L^2(\mu_{>D})} \leq \|g\|_{L^2(\mu)} \leq H. \quad (\text{A.41})$$

Fubini identifies g_D as the density of $x_{>D}$, and Jensen's inequality proves the norm bound. Independence across coordinates ensures that conditioning on c leaves this tail law unchanged. Therefore the conditional law of $(x, y^{(D)})$ has density

$$g_D(x_{>D}) \prod_{i \in I(c)} r_{c,i}(x_i, y_i) \quad (\text{A.42})$$

with respect to $\lambda_{I(c)}$. This follows on cylinder sets from the coordinate construction, and hence on all Borel sets. The rectangle factors form a finite product and are bounded. The tail factor is integrable and need not equal one. Summing the conditional laws over the finitely many patterns proves (A.40).

Proposition A.6 now gives the pair identity β_D -almost everywhere. To pass to the limit, the first marginal of β_D is the x -marginal of ν_j , while its second marginal uses the y -marginals in the first D coordinates and the x -marginals thereafter. Lemma A.7 and (A.37) therefore give uniform absolute continuity of both marginal families, and absolute continuity of the two marginals of ν_j .

On the common probability space, $y^{(D)} \rightarrow y$ in the product topology. Since X^2 is compact metrizable, bounded convergence against continuous functions proves $\beta_D \Rightarrow \nu_j$. Every one of these laws is carried by ordered pairs. Their absolutely continuous marginals put both endpoints in the common conull set on which the cocycles are finite and positive. The ordered-coordinate formula for the cocycle ratios consequently gives $0 < R_{x,y^{(D)}}(k) \leq 1$ and $0 < R_{x,y}(k) \leq 1$, simultaneously for every signed k . All hypotheses of Lemma 3.3 are now satisfied, with the fixed cap $B = 1$. That lemma transfers the identity from β_D to ν_j , proving (A.39). $\square$

A.5 Verifying the convolution obstruction

It remains to exclude invariant projections. We verify the hypotheses of Lemma 5.1 for the product couplings. The infinite-sum estimate is already contained in that lemma; the task here is to bound the finite coefficients on an event of fixed positive probability and to attenuate every omitted index.

The preceding sections have constructed the factor M , its bounded operator T , and all the couplings, with the scales and angles fixed independently of any projection. Let $p \in M$ be an orthogonal projection satisfying $(1 - p)Tp = 0$, and suppose, toward a contradiction, that $0 < p < 1$. Use the common invariant Borel conull set for this fixed p and the cocycles. Equation (A.29) gives

$$0 < r(x) = p(x)_{0,0} < 1 \quad \text{for } \mu\text{-almost every } x.$$

Every coupling used below has absolutely continuous endpoint marginals, so its endpoints belong to this set almost surely. All pointwise operator and cocycle statements below are understood on these sets. Only countably many indices occur, so they may be required simultaneously.

Convolution identities. Set

$$\xi(x) = p(x)e_0, \quad \gamma(y) = (1 - p(y))e_0.$$

These measurable vectors have norms at most one, with $\xi_0(x) = r(x)$ and $\gamma_0(y) = 1 - r(y)$. Fix j . By Proposition A.8, the vector

$$\zeta = \text{diag}_{v \in \mathbb{Z}}(R_{x,y}(v))\xi(x)$$

belongs to $\text{ran } p(y)$ for ν_j -almost every (x, y) . The same range is invariant under the fiber $T(y)$ and therefore under all its nonnegative powers. By the cocycle identity, the coordinate $n + v$ of $T(y)^n \zeta$ is

$$F_n(\sigma^v y) R_{x,y}(v) \xi_v(x) = \frac{F_{n+v}(y)}{F_v(x)} \xi_v(x), \quad n \geq 0.$$

Taking its inner product with $\gamma(y)$ yields

$$0 = \sum_{v \in \mathbb{Z}} \frac{F_{n+v}(y)}{F_v(x)} \xi_v(x) \overline{\gamma_{n+v}(y)} = \sum_{v \in \mathbb{Z}} A_v B_{-n-v}, \quad n \geq 0, \quad (\text{A.43})$$

where

$$A_v = \frac{\xi_v(x)}{F_v(x)}, \quad B_v = F_{-v}(y) \overline{\gamma_{-v}(y)}. \quad (\text{A.44})$$

These sequences themselves need not be square summable. Every convolution in (A.43), however, is absolutely convergent. Indeed $F_{k+1}(z) = f(\sigma^k z) F_k(z) \leq F_k(z)$ for every integer k , so for $n \geq 0$,

$$0 < \frac{F_{n+v}(y)}{F_v(x)} \leq \min\{R_{x,y}(v), R_{x,y}(n+v)\} \leq 1. \quad (\text{A.45})$$

The first bound uses $F_{n+v}(y) \leq F_v(y)$ and the second uses $F_v(x) \geq F_{n+v}(x)$. Cauchy–Schwarz now bounds the absolute sum by $\|\xi(x)\|_2 \|\gamma(y)\|_2 \leq 1$.

An event of uniformly positive probability. Write $K = K_j$ and $m = m_j$. On the augmented probability space $(\Omega_j, \mathbb{P}_j)$, define E_j by the following conditions:

1. For every $i \in J_j$, $x_i < 1 - 2/m$ and $y_i > 2/m$, and at least one coordinate in J_j has the tagged label.
2. For every $s > j$ and $i \in J_s$, $x_i < 1 - 2K/N_s$ and $y_i > 2K/N_s$.
3. $r(x) \geq 1/j$ and $1 - r(y) \geq 1/j$.

We claim that there is a constant $c_* > 0$ such that

$$\mathbb{P}_j(E_j) \geq c_* \quad \text{for all sufficiently large } j. \quad (\text{A.46})$$

The constant can be chosen independently of p ; the point beyond which the bound holds may depend on p .

At one coordinate of J_j , a tagged choice has probability $1/m$ and satisfies both of the first condition's inequalities. By the one-coordinate density bound in Lemma A.7, the probability of a nontagged choice satisfying those inequalities is at least

$$1 - \frac{1}{m} - 10\frac{2}{m} - 10\frac{2}{m} = 1 - \frac{41}{m}.$$

The disjoint events specifying exactly one tag and admissible nontagged choices at the other coordinates consequently give the lower bound

$$m\frac{1}{m} \left(1 - \frac{41}{m}\right)^{m-1} = \left(1 - \frac{41}{m}\right)^{m-1} \geq e^{-82}. \quad (\text{A.47})$$

Here $m \geq 100$ and $\log(1-t) \geq -2t$ for $0 \leq t \leq 1/2$. The probability that the second condition fails is at most

$$40K \sum_{s>j} \frac{m_s}{N_s} = 40K \sum_{s>j} \frac{1}{K_s} \leq \frac{80K_j}{K_{j+1}} \rightarrow 0. \quad (\text{A.48})$$

This is a countable union bound with a convergent sum. The intervals involved have length $2K/N_s$; all are less than one by the scale conditions.

The two endpoint losses in the third condition are bounded, using (A.37), by

$$\mathbb{P}_j\{r(x) < 1/j\} \leq H\mu\{r < 1/j\}^{1/2}, \quad \mathbb{P}_j\{1-r(y) < 1/j\} \leq H\mu\{1-r < 1/j\}^{1/2}.$$

Both tend to zero because $0 < r < 1$ almost everywhere. Subtracting these losses and (A.48) from (A.47) proves (A.46), for example with $c_* = e^{-82}/2$. No independence between these conditions is needed.

The finite coefficients. On E_j the coefficients in (A.44) satisfy

$$|A_v|, |B_v| \leq D_j \quad (|v| \leq K), \quad |A_0|, |B_0| \geq 1/j. \quad (\text{A.49})$$

To verify all signs, first let $0 \leq v \leq K$. For $i \in J_j$, $v\theta_i \leq 2/m$, and for $i \in J_s$ with $s > j$, $v\theta_i \leq 2K/N_s$. The defining endpoint exclusions in E_j therefore give

$$\lfloor x_i + v\theta_i \rfloor = 0, \quad \lfloor y_i - v\theta_i \rfloor = 0 \quad (i \in J_s, s \geq j).$$

At each of the finitely many earlier coordinates, these floors have absolute value at most $K+1$, since $0 < \theta_i < 1$. Thus

$$\frac{1}{F_v(x)} \leq \exp \left((K+1) \sum_{s<j} C_s m_s \right) = D_j, \quad F_{-v}(y) \leq D_j,$$

which bounds A_v and B_v . For the negative coefficient indices, monotonicity gives $F_{-v}(x) \geq 1$ and $F_v(y) \leq 1$, and hence

$$|A_{-v}| \leq 1, \quad |B_{-v}| \leq 1 \quad (0 \leq v \leq K).$$

Finally $A_0 = r(x)$ and $B_0 = 1 - r(y)$, proving (A.49).

Thus the finite-coefficient hypotheses of Lemma 5.1 hold with $D = D_j$ and $a = 1/j$. We next verify its attenuation hypothesis outside this window.

Attenuation at every omitted index. We next control $R_{x,y}(k)$ when $|k| > K$. If $K < |k| \leq N_j/4$, choose a tagged coordinate in J_j on E_j . At that coordinate

$$\frac{1}{m} < |k|\theta_i \leq \frac{1}{2}, \quad x_i < \frac{1}{m}, \quad y_i \geq 1 - \frac{1}{m}.$$

For $k > 0$ the threshold $1 - \{k\theta_i\}$ lies in $[1/2, 1 - 1/m)$; for $k < 0$ it equals $|k|\theta_i$ and lies in $(1/m, 1/2]$. The tagged pair crosses it in both cases. The coordinate contribution to the ratio gives

$$R_{x,y}(k) \leq e^{-C_j} \leq e^{-L_j} \quad \text{on } E_j, \quad K < |k| \leq N_j/4. \quad (\text{A.50})$$

Fix j sufficiently large that the phase conclusions of Proposition A.2 hold in every block $s \geq j$. For $|k| > N_j/4$ we can assign a block J_s and a width w so that either the second phase range is used with $s \geq j$, $w = b_s$, or the first phase range is used with $s > j$, $w = a_s$. Here is the coverage explicitly. Put

$$\alpha_s = G_s K_s, \quad \beta_s = N_s/4, \quad Q_s = m_s^{(4/5)m_s}.$$

The scale conditions imply

$$\alpha_s \leq \beta_s < Q_s, \quad \alpha_{s+1} < Q_s, \quad Q_s \rightarrow \infty.$$

Start with $[\beta_j, Q_j]$, where the second phase range applies. For each $s > j$, the first and second phase ranges together cover $[\alpha_s, Q_s]$, divided at β_s . Consecutive intervals overlap by $\alpha_{s+1} < Q_s$, so their union with the initial interval covers $[\beta_j, \infty)$. These are real intervals; in particular every integer magnitude in question is covered.

Choose one such assignment deterministically for each k , and let $I(k) \subseteq J_s$ be a deterministic list of qualifying coordinates. The choice can, for example, use the least available block and then increasing coordinate order. With $q = |I(k)|$ we have

$$q \geq m_s/20, \quad u_i := 1 - \{k\theta_i\} \in [w, 1 - w] \quad (i \in I(k)). \quad (\text{A.51})$$

All choices depend only on j, k and the already fixed parameters and angles, not on the sampled endpoints. In the second phase range,

$$m_s \rho_s(b_s) = \kappa_s m_s^{1/40} \geq L_s \geq L_j;$$

in the first phase range, where $s > j$,

$$m_s \rho_s(a_s) = \kappa_s \sqrt{G_s} \geq L_{s-1} \geq L_j.$$

Also $C_s \geq L_s \geq L_j$ in either case. If any listed threshold is crossed, then

$$R_{x,y}(k) \leq e^{-C_s} \leq e^{-L_j}. \quad (\text{A.52})$$

Let $\mathcal{B}_k$ be the event that none of the thresholds in (A.51) is crossed, and define

$$a_k(x) = \#\{i \in I(k) : x_i < u_i\},$$

$$H_k^{(1)} = \mathcal{B}_k \cap \{a_k(x) \geq q/2\},$$

$$H_k^{(2)} = \mathcal{B}_k \cap \{a_k(x) < q/2\}.$$

Given x , the eligible coordinates in $H_k^{(1)}$ are fixed. Conditional independence and (A.34) imply

$$\mathbb{P}_j(H_k^{(1)} \mid x) \leq (1 - \rho_s(w)/10)^{q/2} \leq \exp(-m_s \rho_s(w)/400) \leq e^{-L_j/400}. \quad (\text{A.53})$$

For the other orientation, orderedness gives on $\mathcal{B}_k$ the exact equivalence

$$x_i < u_i \iff y_i < u_i \quad (i \in I(k)).$$

Indeed the forward implication follows from the absence of a crossing, and the reverse implication from $x_i \leq y_i$. Thus

$$H_k^{(2)} = \mathcal{B}_k \cap \{\#\{i \in I(k) : y_i \geq u_i\} > q/2\} \quad \text{almost surely.} \quad (\text{A.54})$$

This is the same complementary conditioning split shown in Figure 1, with independent coordinates in place of digit labels. The eligible coordinates in this expression are determined by y . The reverse conditional independence and (A.35) give

$$\mathbb{P}_j(H_k^{(2)} \mid y) \leq e^{-L_j/400}. \quad (\text{A.55})$$

These estimates hold under the full probability law $\mathbb{P}_j$, before any restriction to E_j . There are countably many k , so the conditional estimates can be taken simultaneously outside endpoint null sets. No union estimate over their bad events has been made.

For the near-window indices in (A.50), set both exceptional events equal to the empty set. Combining that estimate with (A.52) and the two conditional bounds, we have, for every $|k| > K_j$,

$$R_{x,y}(k) \leq e^{-L_j} + \mathbf{1}_{H_k^{(1)}} + \mathbf{1}_{H_k^{(2)}} \quad \text{on } E_j,$$

where

$$\mathbb{P}_j(H_k^{(1)} \mid x) \leq e^{-L_j/400}, \quad \mathbb{P}_j(H_k^{(2)} \mid y) \leq e^{-L_j/400}$$

hold under the full law. All inputs to Lemma 5.1 are now verified: the vectors, convolutions and ratio bounds are those of (A.43)–(A.45); the event and finite coefficient bounds are (A.46) and (A.49). For every sufficiently large fixed j , the lemma therefore gives

$$\begin{aligned} c_* \varepsilon_j &\leq (2K_j + 1)(e^{-L_j} + 2e^{-L_j/800}) \\ &\leq (2K_j + 1)3e^{-L_j/10000} \leq \frac{\varepsilon_j}{j}, \end{aligned} \quad (\text{A.56})$$

by (A.2). Since $\varepsilon_j > 0$, cancellation at this fixed stage gives $c_* \leq 1/j$, impossible once $j > 1/c_*$. No uniform lower bound on (ε_j) is used. The scales, angles, operator and coupling laws were fixed before the arbitrary projection p ; only its conull representatives and the eventual threshold in (A.46) depended on p . This excludes every nontrivial invariant projection in M .

A.6 Finite atomic algebras and a diffuse limit

The model also distinguishes the spectral information in finitely many polar conjugates from that in the full orbit. This distinction follows from the explicit weight and the rotation, independently of the convolution argument.

Proposition A.9. *For the operator T constructed above, its polar decomposition is $T = U|T|$, with $|T| = \pi(f)$ and $\ker |T| = \{0\}$. The spectral measure of $|T|$ is countably atomic. For every finite $F \subset \mathbb{Z}$, the abelian von Neumann algebra*

$$\mathcal{A}_F = W^*(U^n |T| U^{-n} : n \in F)$$

is atomic. The full algebra

$$\mathcal{A} = W^*(U^n |T| U^{-n} : n \in \mathbb{Z})$$

is diffuse.

Proof. Since $f > 0$ almost everywhere, $T^*T = \pi(f)^2$ has zero kernel, and $|T| = \pi(f)$. The unitary U is therefore its polar factor. On the conull set where the sum in (A.23) has finite support, the value of f is determined by a finite subset of the countable coordinate set. There are only countably many such subsets. Consequently f takes values in a countable subset of $(0, 1]$, up to a null set. Its nonnull level sets give countably many spectral projections whose sum is 1, which proves the assertion about spectral type.

The representation convention gives

$$U\pi(g)U^* = \pi(g \circ \sigma^{-1}).$$

For a finite F , the tuple $(f \circ \sigma^{-n})_{n \in F}$ has countable range on a common conull set. The nonnull joint level sets therefore give an atomic partition of the generated algebra $\mathcal{A}_F$. This includes $F = \emptyset$, when the generated unital algebra is scalar.

For the infinite family, $\mathcal{A}$ is an invariant abelian subalgebra of $\pi(L^\infty(X))$. The rotation is totally ergodic. Indeed, for any integer $q \neq 0$ and any nonconstant finite-support character indexed by $v = (v_i)$, rational independence implies $q \sum_i v_i \theta_i \notin \mathbb{Z}$. The Fourier argument used in Section A.1 thus proves ergodicity of σ^q .

Suppose e were an atom of $\mathcal{A}$. Its integer translates under $\text{Ad } U$ are atoms of equal positive trace. Distinct atoms in an abelian algebra are orthogonal, so only finitely many can occur. The orbit of e is therefore finite, and for some positive integer q , $U^q e U^{-q} = e$. Because $e \in \pi(L^\infty(X))$, total ergodicity forces $e = 1$. An abelian algebra whose identity is an atom is scalar. But $\mathcal{A}$ contains the nonconstant multiplication operator $\pi(f)$: a positive almost-everywhere constant would have finite logarithmic integral, contrary to Lemma A.5. Hence $\mathcal{A}$ has no atoms, as claimed. $\square$

The factor construction, the norm and determinant calculation, the exclusion of invariant projections, and Proposition A.9 prove Theorem A.1, with $c_i = C_j$ for $i \in J_j$. The diffuseness statement is only about $\mathcal{A}$; it does not identify $\mathcal{A}$ with all of $\pi(L^\infty(X))$.

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